Hamiltonian truncation of the long-range Ising model at finite coupling

Method, window correction, extrapolation, results in $d = 1, 2, 3$, and how to reproduce them

Companion note to the public repository, folder `HT/`

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Abstract

This note documents every finite-coupling Hamiltonian-truncation (HT) number quoted in the paper on the long-range Ising (LRI) model [4]. The perturbative HT coefficients C_2 to C_8 are documented separately, in `HT_Pert/`.

The LRI fixed point is the ϕ^4 deformation of the generalized free field (GFF) [5, 6, 7]. We diagonalize the deformed Hamiltonian in radial quantization, in a truncated Fock space [9, 1], at couplings tuned to the conformal values of two levels. The zero mode costs no level, so the cutoff does not cap its quanta. It is kept in a small basis of states, each a superposition of many of its quanta, and the full Hamiltonian, including the coupling of the zero mode to the bath, is diagonalized on the product of this basis with the bath of all other modes, truncated at a total level N.

In the truncated space an excited bath state can host fewer vacuum fluctuations than the vacuum. A correction for these missing fluctuations, the window correction, removes the dominant descendant-spacing violations of the raw truncation, which do not decay with N. A zero-mode basis selected from a reduced density matrix makes the strong-coupling cutoffs affordable.

In $d = 1$ we reach $N = 40$ at 13 values of Δ_σ and extrapolate in N. The results are final for $\Delta_\sigma \geq 0.1$, one-sided at 0.075, unstable at 0.05, and without a limit for $\Delta_\sigma \leq 0.025$, where the window-corrected model itself does not converge. In $d = 2$ ($N \leq 24$) and $d = 3$ ($N \leq 20$) we report finite-cutoff behaviour only.

Separately written solvers reproduce the finite-cutoff spectra in all three dimensions. In $d = 1$ they agree to 5.6×10^{-7} with the density-basis records behind the quoted values ($N = 12$–20) and to 5×10^{-9} with converged energy-basis records. In $d = 2$ they agree to 1.4×10^{-12} ($N = 12$, 16) and in $d = 3$ to 2×10^{-11} ($N = 8$–12). Section 9 maps each number to its code, records, command and tolerance.

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1 Summary

What is computed. The long-range Ising (LRI) fixed point in d dimensions is reached from the generalized free field (GFF) ϕ of dimension Δ_σ by the relevant ϕ^4 deformation [5, 6, 7, 2]. In radial quantization the GFF becomes a set of free oscillators. We diagonalize the deformed Hamiltonian in a truncated Fock space and tune its two couplings, the mass r and the quartic g, so that two levels take their conformal values: $\sigma = \phi$ at Δ_σ and its shadow at $d - \Delta_\sigma$. The first excited $\mathbb{Z}_2$-even scalar level is then the prediction for the dimension Δ_ε of the energy operator ε (Section 2). The method is Hamiltonian truncation in radial quantization [9, 1], applied to a nonlocal GFF.

The three ingredients.

1. *Zero mode and bath* (Section 2.4). The zero mode costs no level, so its quanta are not capped by the level cutoff, and at strong coupling its fluctuations are large. The full Hamiltonian,

including the couplings of the zero mode to the bath, is diagonalized on the product of n_z zero-mode states per parity, each a superposition of many zero-mode quanta, with the bath states of total level $\ell \leq N$. In $d = 1$ this zero-mode basis reproduces the Δ_ε of a converged zero mode to 10^{-6} (Section 2.5); only the bath cutoff N is extrapolated.

2. *Window correction* (Section 3). A bath state of level ℓ can host vacuum fluctuations only up to level $N-\ell$. The raw spectrum therefore violates the exact descendant spacings by amounts that do not decay with N. We add to every bath state the difference $D(\ell) = E_v(N) - E_v(N-\ell)$, where $E_v(M)$ is the vacuum energy at bath cutoff M of the corrected theory itself, zero mode included. With an energy cutoff, the same missing states were added back order by order with perturbative counterterms in ϕ^4 theory in $2+1$ dimensions [17].

3. *Zero-mode basis* (Section 2.5). If the n_z states are the lowest eigenstates of the zero-mode part H_z of H, the number needed at strong coupling grows with N, and the zero mode then sets the cost. We keep instead the leading eigenvectors of a reduced density matrix of the states the calculation needs, trained once per Δ_σ.

Results.

- In $d = 1$ (Section 4, Table 1) the cutoff reaches $N = 40$ at all 13 values $0.001 \leq \Delta_\sigma \leq 0.249$, and the series are extrapolated with a fixed family of fits.
  - At $\Delta_\sigma = 0.249$ the result differs from the ϵ expansion by -4×10^{-8}, well below the 10^{-6} to which we trust this check.
  - The values are *final* for $\Delta_\sigma \geq 0.1$ as limits of the window-corrected model. Their half-spreads are at most 1.2×10^{-4}; at $\Delta_\sigma \geq 0.15$ we judge the total error to be about 10^{-4}.
  - The quoted errors are half-spreads of the extrapolations in the cutoff, not standard deviations, and they do not include the dependence on the window prescription. That dependence is 7×10^{-4}–2.3×10^{-3} at $0.075 \leq \Delta_\sigma \leq 0.125$, as large as the gap there to the numerical renormalization group (NRG) of the sub-ohmic spin-boson model [11, 12] (Section 4.4).
  - At $\Delta_\sigma = 0.075$ the error is *one-sided*, and at 0.05 the extrapolation is *unstable*.
  - At $\Delta_\sigma \leq 0.025$ the corrected sequence turns over and keeps falling, so it has *no limit*. This is a property of the window-corrected model, not of the numerics.
- In $d = 2$ (Section 5, $N \leq 24$) and $d = 3$ (Section 6, $N \leq 20$) the cutoffs are too few for a controlled extrapolation, and we report finite-N behaviour with status words.

Checks.

- Separately written solvers, built from the definition of the model without using the production code, reproduce the production spectra in each dimension (Section 7). The $d = 1$ and $d = 3$ solvers were written blind to the production code. The $d = 2$ solver is not blind: its author had earlier prepared the production code for the cluster.
- These checks verify that we compute the stated model. Whether its $N \to \infty$ limit is the CFT is a separate question, and the answer is negative in parts of the range (Section 8).

Status words. *Final*, *one-sided*, *unstable*, *no limit* and *finite-N* are part of the results (Section 4.2). A quoted error is a half-spread: half the difference between the largest and the smallest limit of the accepted fits. It is a model spread, not a standard deviation, and it covers the extrapolation only.

2 Model, truncation and tuning

2.1 Radial quantization of the GFF

Radial quantization turns the GFF of dimension Δ_σ into free oscillators a_k of energy $\Delta_\sigma + \kappa_k$:

$$\phi(\hat{n}) = \sum_k u_k \left[f_k(\hat{n})\, a_k + f_k(\hat{n})^* a_k^\dagger \right], \tag{1}$$

$$\begin{aligned}
&d=1: \quad k=n, && \kappa = n, && f = (\pm 1)^n, && u_n^2 = \frac{(2\Delta_\sigma)_n}{n!},\\
&d=2: \quad k=(a,b), && \kappa = a+b, && f = e^{-\mathrm{i}(a-b)\vartheta}, && u_{ab}^2 = \frac{(\Delta_\sigma)_a (\Delta_\sigma)_b}{a!\, b!},\\
&d=3: \quad k=(n,l,m), && \kappa = 2n+l, && f = \sqrt{4\pi}\, Y_{lm}, && u_{nl}^2 = \frac{(\Delta_\sigma)_{n+l} (\Delta_\sigma - \frac{1}{2})_n}{n!\, (\frac{3}{2})_{n+l}},
\end{aligned} \tag{2}$$

These reproduce the two-point function on $\mathbb{R} \times S^{d-1}$; κ_k is the level of mode k and ϑ the angle on the circle. In $d = 1$ the "sphere" S^0 is the pair of points $x = \pm 1$, and $f_n(\pm 1) = (\pm 1)^n$. A Fock state with occupation numbers n_k has:

- level $\ell = \sum_k \kappa_k n_k$;
- particle number $P = \sum_k n_k$;
- free energy $H_0 = \ell + \Delta_\sigma P$.

The zero mode a_0 ($\kappa = 0$, $u_0 = 1$) costs no level.

2.2 Hamiltonian and sectors

The tuning and Δ_ε use the sector of rotation scalars that are even under reflection (inversion in $d = 3$). There we diagonalize

$$H = H_0 + r\, V_2 + g\, V_4, \qquad V_2 = \big\langle :\phi^2: \big\rangle, \qquad V_4 = \tfrac{1}{12} \big\langle :\phi^4: \big\rangle, \tag{3}$$

with $\langle \cdot \rangle$ the average over S^{d-1}. Normal ordering is with respect to all oscillators, the zero mode included. In $d = 1$:

- the reflection $x \to -x$ acts as $a_n \to (-1)^n a_n$, so the reflection-even sector contains the states of even total level;
- on that sector the sphere average equals the vertex at $x = +1$;
- the descendant diagnostic below also diagonalizes the reflection-odd sector (odd total level), which contains the level-1 descendants.

The Ising symmetry $\phi \to -\phi$ splits each sector by the parity of P.

2.3 Tuning conditions and the prediction

We denote by $E_v < E_\varepsilon$ the two lowest $\mathbb{Z}_2$-even scalar levels (the vacuum and the energy operator ε) and by $E_\sigma < E_{\phi^3}$ the two lowest odd ones. We impose

$$E_\sigma - E_v = \Delta_\sigma, \qquad E_{\phi^3} - E_v = d - \Delta_\sigma, \qquad \Delta_\varepsilon = E_\varepsilon - E_v, \tag{4}$$

two conditions for (r, g) and one prediction. The shadow $d - \Delta_\sigma$ is ϕ^3 at weak coupling. The vacuum energy at cutoff N is $E_v = E_v(N)$ in the notation of Section 3. Conformality also requires the descendant spacings

$$v_n(\mathcal{O}) \equiv E(\partial^n \mathcal{O}) - E(\mathcal{O}) - n = 0, \tag{5}$$

a test that needs no continuum input. In $d = 1$ we evaluate v_1 and v_2 by identifying $\partial\mathcal{O}$ through its overlap with $L_{-1}\mathcal{O}$, where L_{-1} is the generator of the free GFF, with the zero-mode operator a_0 projected onto the zero-mode basis of Section 2.5. This is a diagnostic: L_{-1} is not the interacting generator, and ambiguous identifications are flagged.

At weak coupling, $\Delta_\sigma = (d - \epsilon)/4$ with $\epsilon \to 0$:

- the solution has $g = \epsilon/3 + O(\epsilon^2)$ and $r = O(\epsilon^2)$;
- the prediction is $\Delta_\varepsilon = d/2 - \epsilon/6 + C_2(N)\epsilon^2 + \dots$, with the cutoff-dependent coefficients $C_n(N)$ of the truncated model (Section 4.3). Their continuum values are the GFF coefficients C_n, written $C_n(d)$ where the dimension matters.

2.4 Zero mode and bath

We split the field into the zero mode and a bath of all other modes,

$$\phi = q + \chi, \qquad q = a_0 + a_0^\dagger, \qquad p = \mathrm{i}\big(a_0^\dagger - a_0\big), \qquad \chi = \sum_{k\neq 0} u_k \big[f_k\, a_k + f_k^* a_k^\dagger\big]. \tag{6}$$

Normal ordering the zero mode gives $:q^2: = q^2 - 1$, $:q^3: = q^3 - 3q$ and $:q^4: = q^4 - 6q^2 + 3$. Up to a constant,

$$H = H_z \otimes 1 + 1 \otimes H_b + \sum_{j=1}^{4} L_j \otimes W_j, \qquad W_j = \big\langle :\chi^j: \big\rangle, \tag{7}$$

$$H_z = \frac{\Delta_\sigma}{4} p^2 + A\, q^2 + \frac{g}{12} q^4, \qquad A = \frac{\Delta_\sigma}{4} + r - \frac{g}{2}, \tag{8}$$

$$L_1 = (2r - g)\, q + \frac{g}{3} q^3, \qquad L_2 = r - \frac{g}{2} + \frac{g}{2} q^2, \qquad L_3 = \frac{g}{3} q, \qquad L_4 = \frac{g}{12}. \tag{9}$$

- $H_b = \ell + \Delta_\sigma P_b$ is the free bath Hamiltonian.
- In H_z:
  - $\frac{\Delta_\sigma}{4}(p^2 + q^2)$ is the free part of the zero mode;
  - $r q^2$ comes from $r:\phi^2:$;
  - $-\frac{g}{2} q^2$ comes from normal ordering $:q^4:$.
- At the tuned couplings A can be negative, and H_z is then a double well. In $d = 1$, $A < 0$ at every cutoff for $\Delta_\sigma = 0.2$ and 0.175, and its sign changes with the cutoff at $\Delta_\sigma \leq 0.15$.
- L_j is the zero-mode operator that multiplies W_j in $r:\phi^2: + \frac{g}{12}:\phi^4:$. For example, L_1 collects $2rq\chi$ and $\frac{g}{3}:q^3:\chi$.

Truncation. We keep all bath states with level $\ell \leq N$, with no limit on the bath particle number, and n_z zero-mode states per parity (Section 2.5). The full H of Eq. (7), the couplings $L_j \otimes W_j$ included, is diagonalized on their product; the eigenstates of H_z alone serve at most to choose the zero-mode states. The normal-ordered W_j never pass through a level above both endpoints, so H at cutoff $M < N$ is exactly the $\ell \leq M$ projection of H at cutoff N. This property is used by the window correction. The same split is used in $d = 2, 3$, with the $J = 0$ projection of Section 6 in $d = 3$.

2.5 Zero-mode basis

H_z and the L_j are represented in an auxiliary harmonic-oscillator basis of $N_{\rm osc}$ states ($N_{\rm osc} = 384$ in production), with an adaptive scale t. Of this basis we keep n_z states per parity. The product space has $2n_z$ times as many states as the bath, so n_z sets the cost.

Energy basis. The original choice keeps the n_z lowest eigenstates of H_z in each parity. At strong coupling in $d = 1$ this basis is not converged at fixed n_z beyond $N \approx 24$, because the tuned g grows with N. At $\Delta_\sigma = 0.025$, $N = 28$, $n_z = 32$, 40 and 48 give $\Delta_\varepsilon = 0.710632, 0.710934, 0.710959$ (window). The energy-basis campaign `prod1--7` (Section 4.1) therefore needed n_z up to 56.

Density basis. Instead, we keep the zero-mode states that the low-lying spectrum actually uses: the leading eigenvectors, in each parity, of the reduced density matrix

$$\rho = \tfrac{1}{6}\big(\rho_{\rm vac} + \rho_\sigma + \rho_{\phi^3}\big) + \frac{1}{N_t} \sum_{M=0,2,\ldots}^{N_t-2} \rho_{\rm vac}(M). \tag{10}$$

- *What ρ averages.* It is obtained by tracing the bath out of the states the calculation needs. The first term covers the vacuum, σ and ϕ^3, which enter the tuning, at the training cutoff N_t. The sum covers the vacua at all lower even cutoffs, which enter the window correction. The ε state plays no role in the selection.

- *Training.* The basis is trained once per Δ_σ at $N_t = 24$, against a reference calculation in the energy basis with 48 states per parity (the *teacher*). The outer loop alternates density selection and retuning in the selected span. It stops when the couplings change by less than 2×10^{-8} and the teacher's tuning residuals are below 10^{-7}, after at least two sweeps.

- *Stored basis.* Each trained basis is stored in a file that keeps 24 states per parity: the isometry U and the scale t. Its first n_z columns give a nested smaller basis, and production uses $n_z = 16$.

- *Operators.* The powers q^j are formed in the full auxiliary basis before projection, $q_j = U^{\mathsf{T}}(t^j Q^j)U$; powers of the projected q are not used. For the descendant diagnostic the actual a_0 is projected.

- *What is frozen.* The span and t are fixed within a solve, while the window table is recomputed at every coupling.

In $d = 1$ the density basis reproduces the Δ_ε of a converged zero mode to 10^{-6} at a tenth of the cost. Two comparisons support this:

- against the $n_z = 48$ and 56 energy-basis records at $N = 32$: at most 6.4×10^{-7};

- against the blind solver's grid zero mode at $N \leq 20$: at most 5.6×10^{-7}.

The difference between 24 and 16 kept states is at most 1.8×10^{-9} at $N = 36$, 40. That difference measures only the truncation within one span.

2.6 Tuning

The couplings are tuned by Newton's method on Eq. (4), with backtracking: the Newton step is halved until the norm of the two residuals decreases, at most 13 times. The Jacobian comes from Hellmann–Feynman derivatives of the levels; with the window it includes the window response, Eq. (13). In $d = 1$ the internal coordinates are $\theta = (\theta_0, \theta_1)$, with $g = 192e^{3\theta_1}/\Delta_\sigma^2$ and $A = 4e^{2\theta_1}\theta_0/\Delta_\sigma$ (as in $d = 2$, Section 5). Points are continued from weak coupling or from saved seeds; a seed is only a starting point.

Tolerances:

- tuning residuals $\leq 5 \times 10^{-10}$;
- ARPACK tolerance 3×10^{-11}, eigenvector residuals $\leq 2 \times 10^{-7}$;
- window fixed points $\leq 10^{-12}$.

3 The window correction

The truncation artifact. In the CFT every descendant spacing $E(\partial^n \mathcal{O}) - E(\mathcal{O}) = n$ is exact. The raw truncation violates these spacings by amounts that do not decay with N. In $d = 1$ the level-1 violations are 0.06–0.25 at $\Delta_\sigma \leq 0.1$ over $N = 16$–28 (Table 3 shows $N = 28$). The mechanism is that a bath state of level ℓ can carry vacuum fluctuations of the bath only up to level $N - \ell$, whereas the vacuum keeps all fluctuations up to N. The same mechanism spoils the cancellation of disconnected vacuum diagrams in ϕ^4 theory in $2 + 1$ dimensions with an energy cutoff, where the effect does not decouple as the cutoff is removed; there it was addressed with perturbative state-dependent counterterms [17]. The missing vacuum window is the truncation artifact; in the perturbative truncation it is analysed in `HT_Pert/`.

The correction. We add to every bath state of level ℓ, in all sectors, the shift

$$D(\ell) = E_v(N) - E_v(N - \ell), \qquad D(0) = 0, \tag{11}$$

where $E_v(M)$ is the vacuum energy at bath cutoff M of the same Hamiltonian, with the same Δ_σ, couplings, zero mode and basis. The definition of E_v has three features:

- *Zero mode included.* Vacua of the bath alone miss the zero-mode tadpole $\frac{g}{2}\langle q^2 \rangle$, which cancels the $-\frac{g}{2}$ in A, and remove only part of the violation.
- *Consistent.* The production mode is *consistent*: $E_v(M)$ is the lowest eigenvalue of the *corrected* Hamiltonian at cutoff M,

  $$E_v(M) = \lambda_{\min}\big[H(M) + D_M\big], \qquad D_M(\ell) = E_v(M) - E_v(M - \ell), \tag{12}$$

  built level by level from the corrected lower cutoffs.
- *Odd cutoffs.* E_v of an odd M equals $E_v(M - 1)$, since only even levels exist in the sector.

The *literal* mode uses the uncorrected $\lambda_{\min} H(M)$ and serves as a diagnostic. Energies are measured from the lowest even eigenvalue of the zero mode alone, so $E_v(0) = 0$.

Precedent. $D(\ell)$ is the counterpart, with a level cutoff, of the state-dependent counterterms that Ref. [17] adds with an energy cutoff to restore the missing states of disconnected bubbles.

- At second order in the interaction, $D(\ell)$ corresponds to their single-bubble counterterm (Patch I, their Eq. (5.38)). Their denominators contain the unperturbed energy of the level being computed; ours contain the bubble energy alone.
- They add such terms order by order, including towers of repeated bubbles (Patch II, up to five bubbles), as a pragmatic fix. $D(\ell)$ is a nonperturbative analogue: it takes all disconnected bubbles at once from the vacuum energies of the same truncated Hamiltonian at lower cutoffs. The consistent definition, Eq. (12), also corrects the states inside those vacuum energies, which is the role of their Patch II.

Fixed point. At each M, Eq. (12) is a scalar fixed point $s = \lambda_{\min}[H(M) + D_M(s)]$, because $s = E_v(M)$ enters every $D_M(\ell > 0)$. We write $F(s) = \lambda_{\min} - s$. Then $F'(s) = -p_0$, with p_0 the weight of the vacuum vector on bath level 0. F is concave, so Newton's method converges monotonically after at most one step, and the fixed points are solved to 10^{-12}. At $M = N$, s is the ground energy of the main problem.

Derivatives. For the tuning Jacobian, Hellmann–Feynman on the stored vacuum vectors gives

$$\partial E_v(M) = \frac{\langle \partial H \rangle - \sum_{\ell>0} w_\ell \, \partial E_v(M-\ell)}{p_0}, \tag{13}$$

with w_ℓ the weight of the vacuum vector on bath level ℓ. The package tests compare it with finite differences.

What it removes and what it misses.

- In $d = 1$ the correction reduces the level-1 violation of σ by a factor 10–40 at $N = 28$, $\Delta_\sigma \leq 0.15$.
- At $\epsilon \geq 0.6$ ($\Delta_\sigma \leq 0.1$), non-decaying residuals remain:
  - about 0.01 for σ and ϕ^3;
  - up to 0.05 for ε at $\Delta_\sigma \leq 0.05$ (Table 3).
- The correction restores only disconnected vacuum fluctuations. At finite coupling it therefore models the truncation artifact rather than removing it completely, and the residuals show what it misses.
- Expanded in ϵ, the corrected $d = 1$ model changes Δ_ε only at $O(\epsilon^4)$: in the blind solver the difference to the raw Δ_ε scales as ϵ^4 (Section 7). C_2 and C_3 are therefore unaffected.
- The perturbative windows of `HT_Pert/` are built in the same spirit. In $d = 1$ the finite-coupling window has not been compared with them order by order, so whether its $O(\epsilon^4)$ term is the one the continuum requires is not known.
- In $d = 2, 3$ the consistent window is the same construction (Sections 5, 6). In $d = 2$ its ϵ expansion reproduces the consistent-window scheme of `HT_Pert/`, whose fourth-order coefficient grows like $\ln N$; the model is therefore correct only through third order. In $d = 3$ its coefficients $C_n(N)$ come from `HT_Pert/`: C_2 and C_3 approach their continuum values, while the fourth-order coefficient at $N = 20$ is 60% of $C_4(3)$ and still growing slowly, a finite-N statement.

We have not constructed an effective Hamiltonian beyond the window [15, 16].

4 One dimension

4.1 Campaigns

All $d = 1$ runs used the solver `HT/code/lri_dressed_phi4_hpc` on the symmetry cluster (Perimeter Institute). There were two campaigns.

`dens_v1` (production, code 33998bb, 2026-10-07 to 10-09).

- *Setup.* Consistent window, density basis with $n_z = 16$ (one stored basis per Δ_σ, trained at $N_t = 24$), auxiliary oscillator basis of $N_{\mathrm{osc}} = 384$ states.
- *Grid.* The 13 values $\Delta_\sigma = 0.249$, 0.24, 0.225, 0.2, 0.175, 0.15, 0.125, 0.1, 0.075, 0.05, 0.025, 0.01 and 0.001, that is $\epsilon = 1 - 4\Delta_\sigma$ from 0.004 to 0.996, at every even N from 12 to 40: 195 points.
- *Controls.* Six controls with 24 kept states at $N = 36$, 40. Two further controls, at $N = 40$ and $\Delta_\sigma = 0.025$, 0.01, stopped at their walltime and are not used.
- *Execution.* Items were packed on whole nodes (64-core AMD amdq; 40-core Intel preq), with 1 to 16 threads per item. Items resumed from Newton checkpoints after a cluster-wide node failure and after walltime stops.
- *Cost.* The longest items took 15 h ($N = 40$, $\Delta_\sigma = 0.025$, 16 threads). The total was about 10^4 core-hours of exclusive nodes. Every record carries its configuration, couplings, levels, window table, residuals and timing, and a sha256 over its content.

`prod1--7` (code 4dfd833, 2026-10-02 to 10-05).

- *Content.* Energy zero-mode basis with $n_z = 24$ to 56, in the raw, consistent and literal prescriptions, at $N = 12$ to 40: 680 records, about 1.06×10^4 core-hours.
- *Why it was superseded.* At $\Delta_\sigma \le 0.05$ its series needed an n_z that grows with N, and their extrapolations mixed values of different n_z. The production values now come from `dens_v1`.
- *What is still used.* Its records serve three roles: the inputs of the descendant-identity diagnostic (Section 4.5), the references of the basis check at $N = 32$, and the references of the blind-solver comparison.

4.2 Extrapolation

Fits. For each Δ_σ we fit

$$\Delta_\varepsilon(N) = \Delta_\infty + aN^{-\omega} \quad \big[+ bN^{-\omega-1} \text{ (F2) or } + bN^{-2\omega} \text{ (F3)} \big], \qquad N \ge N_{\min}, \tag{14}$$

with F1 the two-term form. The fit set is:

- $N_{\min} = 12$–24 in steps of 2, giving 21 fits per Δ_σ;
- each fit is a nonlinear least-squares fit from five starting exponents, keeping the best.

Acceptance. A fit is accepted if all three conditions hold:

- rms residual $< 3 \times 10^{-6}$;
- $0.2 < \omega < 4$;
- $|\Delta_\infty - \Delta_\varepsilon(N_{\max})| < 0.05$.

We quote the median of the accepted Δ_∞ and their half-spread, half the difference between the largest and the smallest accepted Δ_∞. It is a model spread, not a standard deviation.

Stability. A result is *stable* if all three conditions hold:

- at least five fits are accepted;
- the median of their fitted ω is at least 0.45 (a condition on the fits, not on ω_{eff} below);
- the half-spread is at most 2×10^{-3}.

The median exponents of the stable results lie at 0.6–0.9.

Status words. A stable result is *final* if the local exponent ω_{eff} below levels off over the largest cutoffs, and *one-sided* if it still rises, which suggests that the fits overestimate the limit; this classification is a judgement. A series that turns over and keeps falling has *no limit*. Any other result that is not stable is *unstable*: we give the range of the accepted fits but no limit. In $d = 2, 3$, *finite-N* marks values of the truncated model at a given cutoff, with no extrapolation.

Local exponent. A fit-free check is

$$\omega_{\text{eff}}(N) = -\frac{d \ln \left(d\Delta_\varepsilon / d \ln N \right)}{d \ln N}, \tag{15}$$

which is constant for a single power law. We evaluate it by finite differences. The slope on a pair of successive cutoffs, $s = [\Delta_\varepsilon(N'+2) - \Delta_\varepsilon(N')]/\ln \frac{N'+2}{N'}$, sits at $\bar{N} = \sqrt{N'(N'+2)}$. From the slopes on $[N', N'+2]$ and $[N'+4, N'+6]$ we take $\omega_{\text{eff}} = -\ln(s_2/s_1)/\ln(\bar{N}_2/\bar{N}_1)$ and label it by the centre $N = N'+3$ of the four cutoffs, $N = 15, 19, \ldots, 35$.

Held-out test. Fits restricted to $N \le N_{\text{cut}}$ predict the cutoffs above N_{cut} (Table 2).

4.3 Results

Table 1 gives Δ_ε at the largest cutoff and extrapolated. Figure 1 shows the strong-coupling series and their local exponents.

Weak coupling and the ϵ expansion.

- At $\Delta_\sigma = 0.249$ the extrapolation differs from the ϵ series through ϵ^7 (GFF folder: $C_2 = (\pi + 4\ln 2)/9$, C_3 to C_7) by -4×10^{-8}. That is below the 10^{-6} at which we trust the check, because of the series and fit systematics.
- At $\Delta_\sigma = 0.24$ the value lies within the band 0.49419–0.49426 of the partial sums through ϵ^4 to ϵ^7.

Δ_σ	ϵ	$\Delta_\varepsilon(N=40)$	$\Delta_\varepsilon(N\to\infty)$	status	NRG
0.249	0.004	0.4993430	0.49934358(9)	final	0.508030^{nc}
0.24	0.04	0.494186	0.4942198(26)	final	0.498922^{nc}
0.225	0.1	0.487982	0.488080(5)	final	0.489668^{nc}
0.2	0.2	0.482839	0.483024(5)	final	0.482980(708)
0.175	0.3	0.483541	0.483909(15)	final	0.483900(346)
0.15	0.4	0.490388	0.491209(33)	final	0.491106(115)
0.125	0.5	0.504561	0.50612(4)	final	0.505672(54)
0.1	0.6	0.528210	0.53048(12)	final	0.529608(33)
0.075	0.7	0.564851	0.56733(20)(−50)	one-sided	0.566260(30)
0.05	0.8	0.620671	0.6220–0.6224, or down to 0.618	unstable	0.622115(27)
0.025	0.9	0.710577		no limit	0.713422(29)
0.01	0.96	0.809111		no limit	0.809747(47)
0.001	0.996	0.969338		no limit	0.937424(111)

Table 1: Δ_ε in $d=1$ from finite-coupling HT, at the largest cutoff and extrapolated with Eq. (14). Errors in parentheses are half-spreads of the accepted fits (model spreads, not standard deviations), printed with the paper's rounding rule. *Weak coupling.* At $\Delta_\sigma \geq 0.15$ these spreads, at most 3.3×10^{-5}, are probably optimistic; we judge the total error there to be about 10^{-4}. At $\Delta_\sigma = 0.249$ this is conservative: the ϵ expansion agrees to -4×10^{-8} (Section 4.3). $\Delta_\sigma = 0.075$. The second error is one-sided: a further rise of ω_{eff} by 0.2 would lower the limit by about 5×10^{-4}. $\Delta_\sigma = 0.05$. The accepted F1 and F2 fits give the range shown, an upper estimate if ω_{eff} keeps rising. The F3 fits admit a later turnover and reach the lowest value shown. The data up to $N=40$ cannot decide, so we quote no limit there, only this range, and none at $\Delta_\sigma \leq 0.025$. The errors do not include the dependence on the window prescription (Section 4.4). *NRG.* Bosonic NRG of the sub-ohmic spin-boson model [11, 12] (NRG folder, final table), shown for comparison. Its error is the sum of three convergence diagnostics (chain depth, kept states, discretization), not a standard deviation; ^{nc}: not converged in chain depth near $\Delta_\sigma = \frac{1}{4}$.

- The blind solver of Section 7 computes the weak-coupling coefficient of the truncated model. With $\Delta_\varepsilon = \frac{1}{2} - \frac{\epsilon}{6} + C_2(N)\epsilon^2 + \ldots$, it reproduces

$$C_2(N) = \frac{34}{81} + \sum_{n \text{ even},\, 2\leq n\leq N} v_n\Big[\frac{2}{81(n-\frac{1}{2})} + \frac{2}{9n} + \frac{2}{9(n+\frac{1}{2})} + \frac{2}{81(n+1)}\Big], \qquad v_n = \frac{(\frac{1}{2})_n}{n!}, \tag{16}$$

  to at most 3×10^{-7} for $N=0$ to 16, raw and corrected. The limit of $C_2(N)$ is the GFF value; the approach is like $N^{-1/2}$.

Intermediate coupling.

- The extrapolations are final for $\Delta_\sigma \geq 0.1$.
- At $0.075 \leq \Delta_\sigma \leq 0.125$ they add 1.6×10^{-3}–2.5×10^{-3} to the $N=40$ values.
- Over $15 \leq N \leq 35$ the local exponent levels off:
  - at 0.64 for $\Delta_\sigma = 0.125$;
  - at 0.68 for 0.1.

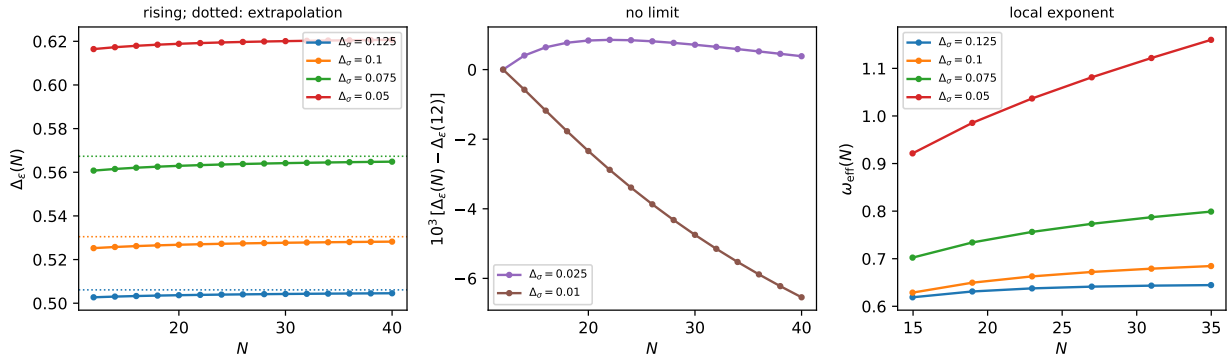


Figure 1: $d = 1$, `dens_v1`. *Left:* $\Delta_\varepsilon(N)$ at $0.05 \le \Delta_\sigma \le 0.125$; the dotted lines are the extrapolations of the stable series (0.075 to 0.125). *Middle:* change $\Delta_\varepsilon(N) - \Delta_\varepsilon(12)$ at $\Delta_\sigma = 0.025$, which peaks near $N = 22$ and then falls, and at 0.01, which falls at a nearly constant rate per unit of $\ln N$. *Right:* the local exponent (15), which levels off at 0.125 and 0.1 but rises at 0.075 and 0.05.

Δ_σ	$\omega_{\rm eff}(15)$	(19)	(23)	(27)	(31)	(35)
0.249	0.39	0.42	0.44	0.45	0.46	0.46
0.24	0.50	0.53	0.54	0.55	0.56	0.57
0.2	0.78	0.79	0.80	0.80	0.81	0.81
0.15	0.67	0.67	0.68	0.67	0.67	0.67
0.125	0.62	0.63	0.64	0.64	0.64	0.64
0.1	0.63	0.65	0.66	0.67	0.68	0.68
0.075	0.70	0.73	0.76	0.77	0.79	0.80
0.05	0.92	0.99	1.04	1.08	1.12	1.16
0.025	3.19	8.09	slope changes sign near $N = 22$			

Table 2: Local exponent (15) of the $d = 1$ series, labelled by the centre of its four cutoffs. Held-out test: fits on $N \le 32$ predict the values at $N = 34$–40 to within 2.3×10^{-6} at every $\Delta_\sigma \ge 0.05$, inside the spread of the predictions. The test passes also at $\Delta_\sigma = 0.025$, where the series has no limit, so it checks the fits only within the range of the data, not their limit. All values: `results/HT_1d/holdout_and_local_exponent.md`.

- It rises instead, over the same range of N:
  - 0.70–0.80 at 0.075;
  - 0.92–1.16 at 0.05.
- A rising $\omega_{\rm eff}$ precedes the turnover seen at $\Delta_\sigma = 0.025$ and suggests that the fits overestimate the limit. This is why 0.075 carries a one-sided error. The classification final versus one-sided is a judgement based on these trends; they differ only modestly between 0.1 and 0.075.
- At weak coupling $\omega_{\rm eff}$ also rises slowly (0.39–0.46 at $\Delta_\sigma = 0.249$), but the bias it implies is below 10^{-5}.
- Fits restricted to $N \le 32$ predict $N = 34$–40 to 2.3×10^{-6} (Table 2).

Strong coupling.

- $\Delta_\sigma = 0.05$. The series still rises at $N = 40$. The accepted F1 and F2 fits give 0.6220–0.6224, a range that contains the NRG value of Table 1. The F3 fits reach down to 0.618. Their low branch moves up as cutoffs are added, which mildly argues against a turnover. The rising ω_{eff} argues the other way. No value is quoted.

- $\Delta_\sigma \leq 0.025$. The corrected sequence does not converge.

  – At 0.025 it peaks near $N = 22$ and then falls ever faster, reaching -1.3×10^{-3} per unit of $\ln N$ at $N = 40$.

  – At 0.01 it falls at a constant -6.2×10^{-3} per unit of $\ln N$.

  – At 0.001 it falls slowly but lies 0.032 above the NRG value.

  The zero-mode basis is converged and the solver is verified, so this is a property of the window model, not of the numerics. The uncorrected truncation converges, but to values that violate Eq. (5) (Table 3).

4.4 Model dependence and the comparison with NRG

The errors of Table 1 are half-spreads of one family of fits (F1–F3) on one prescription. They do not cover two further model dependences, which we measure on the shipped records (`public/model_dependence_ht1d.py`, table in `results/HT_1d/model_dependence.md`).

- *Tail model.* We tried two alternative descriptions of the approach $N \to \infty$ of the same window series:

  – F4, Eq. (14) with two free exponents;

  – a local exponent continued linearly in $\ln N$ from its last four values, with the slope at $N = 40$ integrated to infinity.

  Both move the limit down by at most 1.6×10^{-4} at $0.075 \leq \Delta_\sigma \leq 0.125$, of the order of the fit half-spread.

- *Prescription.* The same truncation without the window (raw, energy zero-mode basis) extrapolates, with the same fits, to limits $6.6 \times 10^{-4}, 2.3 \times 10^{-3}, 1.7 \times 10^{-3}$ below the window limits at $\Delta_\sigma = 0.125$, 0.1 and 0.075, respectively.

  – Neither prescription satisfies the descendant identities in the limit (Table 3), so the limit of the truncated model depends on the prescription at this level.

  – At $\Delta_\sigma \geq 0.15$ the two prescriptions agree to 1.3×10^{-4} or better.

The prescription dependence, 7×10^{-4}–2.3×10^{-3} at $0.075 \leq \Delta_\sigma \leq 0.125$, is thus as large as the gap to NRG below. We do not combine these shifts into an error. The truncation has no strict error bar, and an error built after the gap to NRG had been seen would make the comparison read as agreement within errors. The errors of Table 1 stay the fit spreads. The window limit is expected to be the better of the two, since its σ descendant violation is 10–40 times smaller.

Comparison with NRG. We compare with the final NRG values of Table 1 (NRG folder). Their errors are sums of three convergence diagnostics, not standard deviations.

- HT lies above NRG at all six points $0.075 \le \Delta_\sigma \le 0.2$, by up to 1.1×10^{-3}.
- The differences are 10^{-4} to 1.1×10^{-3} at $0.075 \le \Delta_\sigma \le 0.15$, and below 10^{-4} at 0.175 and 0.2.
- At $\Delta_\sigma = 0.125$, 0.1 and 0.075 the differences are $4.5 \times 10^{-4}, 8.7 \times 10^{-4}, 1.1 \times 10^{-3}$. They are at least five times both the fit half-spreads of Table 1 and the NRG errors, and of the size of the prescription dependence.
- NRG lies between the raw and the window limits at these three points.
- NRG also lies inside the fit-free bracket $[\Delta_\varepsilon(40),\ \Delta_\varepsilon(40) + \frac{d\Delta_\varepsilon/d\ln N}{\omega_{\rm eff}}]$ of the window series at every $0.075 \le \Delta_\sigma \le 0.2$. That bracket holds if $\Delta_\varepsilon(N)$ keeps rising and $\omega_{\rm eff}$ does not decrease beyond $N = 40$.

We do not assume that either method is right. The Monte Carlo values that the MC folder labels *accepted* (MC column of the $d = 1$ results table of Ref. [4]) also lie above NRG at these three couplings, on the same side as HT, but their systematic errors exceed the HT–NRG differences, so they cannot decide between the two. The resummed expansions cannot decide at this coupling (`Resummation/` folder). The HT values at $0.075 \le \Delta_\sigma \le 0.125$ are limits of the window-corrected truncation at their fit errors. As estimates of the CFT they depend on the truncation prescription by the shifts above, which are as large as the gap to NRG.

4.5 Descendant identities

Table 3 gives the level-1 violations, Eq. (5), at $N = 28$, raw and with the window, from the identity diagnostic run on the `prod1--7` records. The other cutoffs are in `results/HT_1d/identity_violations.md`.

- The raw violations do not decay with N over $N = 16$–28 and grow with the coupling. At $\Delta_\sigma \le 0.1$ they range over 0.06–0.25.
- The window reduces the σ violation by 10–40 times at $\Delta_\sigma \le 0.15$.
- At $\epsilon \ge 0.6$ non-decaying residuals of about 0.01 remain for σ and ϕ^3, and up to 0.05 for ε at $\Delta_\sigma \le 0.05$.

5 Two dimensions

5.1 Modes, sectors and couplings

In $d = 2$ the exponent s of the long-range interaction fixes $\Delta_\sigma = (d - s)/2 = 1 - s/2$, so $\epsilon = 2s - 2 = 2 - 4\Delta_\sigma$. The interacting window is $1 < s < s_*$, that is $\frac{1}{8} < \Delta_\sigma < \frac{1}{2}$. Its upper end is the crossover to the short-range Ising CFT, at $s_* = 2 - \eta_{\rm SR} = \frac{7}{4}$ [6, 8, 2]. At $s = s_*$ the fixed point is the short-range Ising CFT decoupled from a generalized free field [8, 2], so there $\Delta_\varepsilon = 1$, the dimension of the energy operator of the two-dimensional Ising model [13]. The scan covers $\frac{1}{8} \le \Delta_\sigma < \frac{1}{2}$.

The modes are $k = (a, b)$, $a, b \ge 0$, of Eq. (2): the oscillator a_{ab} has level $\kappa = a + b$, spin $j = a - b$ and weight $u_{ab}^2 = (\Delta_\sigma)_a (\Delta_\sigma)_b / (a!\, b!)$. With $f_{ab} = e^{-{\rm i}(a-b)\vartheta}$, the two-point function of the field (1) is the cylinder covariance $z^{\Delta_\sigma} |1 - z\, e^{{\rm i}\vartheta}|^{-2\Delta_\sigma}$, with $z = e^{-\tau}$ the radial-time variable and ϑ the angle on

	raw			window		
Δ_σ	σ	ϕ^3	ε	σ	ϕ^3	ε
0.249	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000
0.2	+0.0030	+0.0067	+0.0048	+0.0007	+0.0044	+0.0024
0.15	+0.0207	+0.0297	+0.0238	+0.0007	+0.0096	+0.0040
0.125	+0.0388	+0.0492	+0.0393	−0.0011	+0.0088	−0.0003
0.1	+0.0644	+0.0743	+0.0580	−0.0044	+0.0046	−0.0100
0.075	+0.0990	+0.1056	+0.0797	−0.0088	−0.0024	−0.0248
0.05	+0.1481	+0.1439	+0.1040	−0.0123	−0.0110	−0.0423
0.025	$+0.2321^a$	+0.1869	+0.1273	-0.0063^a	−0.0186	−0.0548
0.01	$+0.3483^a$	+0.2062	+0.1320	$+0.0345^a$	−0.0201	-0.0519^a

Table 3: Level-1 descendant violations v_1 at $N = 28$, raw and with the consistent window, in the energy zero-mode basis of `prod1--7`: $n_z = 24$ at $\Delta_\sigma \geq 0.1$, 32 below. The ε columns use the first excited even state. ^{a}: ambiguous identification with the free L_{-1}. In the CFT all entries vanish.

the circle. The oscillator a_{00} is the zero mode of Eq. (6), and all others form the bath. A bath state with occupations n_{ab} has level, particle number and spin

$$\ell = \sum_{(a,b)\neq(0,0)} (a+b)\, n_{ab}, \qquad P = \sum n_{ab}, \qquad J = \sum (a-b)\, n_{ab}. \tag{17}$$

We keep every bath state with $\ell \leq N$, with no limit on P (Section 2.4). The scalar spectrum is computed at $J = 0$, in the sector even under the reflection $a \leftrightarrow b$. Since $\ell - J = 2\sum b\, n_{ab}$ is even, every scalar bath state has even level. The truncations at $N = 2m$ and $2m+1$ therefore coincide, and we use the even cutoffs $N = 8, 10, \ldots, 24$.

The vertices $W_j = \langle{:}\chi^j{:}\rangle$ of Eq. (7) are built exactly in the retained scalar space. With $B = \sum_{(a,b)\neq(0,0)} u_{ab}\, a_{ab}$ the annihilation part of χ at $\vartheta = 0$, and S the embedding of the scalar space,

$$W_j = \sum_{i=0}^{j} \binom{j}{i} (B^i S)^\dagger (B^{j-i} S). \tag{18}$$

Between two scalar states the average over the circle equals the value at $\vartheta = 0$, so no further projection is needed; states of nonzero spin enter as intermediate states. Since B lowers the level, $B^{j-i}S$ stays inside the space $\ell \leq N$, and the matrix elements are exact: no intermediate state is cut off.

The couplings are parametrized by $\theta = (\theta_0, \theta_1)$,

$$g = \frac{192\, e^{3\theta_1}}{\Delta_\sigma^2}, \qquad A = \frac{4\, e^{2\theta_1}\theta_0}{\Delta_\sigma}, \qquad r = A + \frac{g}{2} - \frac{\Delta_\sigma}{4}, \tag{19}$$

with A the coefficient of q^2 in H_z, Eq. (8). The two tuning conditions of Eq. (4), $E_\sigma - E_v = \Delta_\sigma$ and $E_{\phi^3} - E_v = 2 - \Delta_\sigma$, fix θ, and the prediction is $\Delta_\varepsilon = E_\varepsilon - E_v$.

5.2 Zero-mode basis and window

The zero-mode basis is the density basis of Section 2.5, Eq. (10), with the training cutoff $N_t = 6$. The teacher is an energy-basis calculation at $N = 6$ with 24 states per parity, taken from an auxiliary oscillator basis of $N_{\text{osc}} = 192$ states. Each stored basis keeps the 20 leading eigenvectors of ρ per

parity, which contain the nested bases of $n_z = 12$, 16 and 20 states. Production uses $n_z = 16$, and $n_z = 20$ and 12 are recomputed at every cutoff as checks. For ten values of Δ_σ the stored bases are those of the package; for $\Delta_\sigma = 0.125$, 0.135 and 0.175 new ones were trained with the same settings. No basis is transferred between values of Δ_σ. Over all chains and cutoffs,

$$|\Delta_\varepsilon(n_z{=}20) - \Delta_\varepsilon(n_z{=}16)| \le 9.4 \times 10^{-11}, \qquad |\Delta_\varepsilon(n_z{=}12) - \Delta_\varepsilon(n_z{=}16)| \le 5.2 \times 10^{-7},$$

the latter at $\Delta_\sigma = 0.125$, $N = 22$. Three reference points at $N = 14$ ($\Delta_\sigma = 0.126$, 0.25 and 0.4) use instead the energy basis with 32 states per parity, recomputed at every coupling from $N_{\mathrm{osc}} = 256$; they agree with $n_z = 16$ to 10^{-9}.

The window is the consistent correction of Section 3, Eqs. (11) and (12). Every bath state of level ℓ is shifted by $D(\ell) = E_v(N) - E_v(N - \ell)$. The vacua $E_v(M)$, $M = 0, 2, \ldots, N$, are computed with their own windows, at the same couplings and in the same zero-mode basis; they are not retuned. Their fixed points are solved as in Section 3 and accepted at 10^{-11}, and the derivatives (13) enter the tuning Jacobian. The other acceptance thresholds are a tuning residual $\le 1.5 \times 10^{-10}$, an eigenvector residual $\le 10^{-8}$ and an orthogonality defect $\le 10^{-9}$. Every shipped record passes them (check `d2.records.gates`).

5.3 Campaign

Thirteen chains, $\Delta_\sigma \in \{0.125, 0.126, 0.135, 0.15, 0.175, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.475, 0.495\}$, ran from $N = 8$ to 24 in steps of two on Expanse (SDSC), one chain per 64-core, 64 GB allocation. Each cutoff was seeded by the previous one, and the first by the package's $N = 8$ reference solution or by the teacher. The 348 records (main runs, basis checks and reference points) took 1.1×10^4 core-hours; a point at $N = 24$ took 0.6–11 h and up to 24 GiB. The code is `lri_2d_window_hpc` at commit 889a968, and every record carries a hash of that source. The Python environment was frozen (Python 3.11.5, NumPy 2.3.5, SciPy 1.17.0, Numba 0.65.1). The lowest value, $\Delta_\sigma = 0.125$, is the crossover point $s = s_*$, where $\Delta_\varepsilon = 1$.

5.4 Finite-cutoff behaviour

Table 4 and Fig. 2 give $\Delta_\varepsilon(N)$. All of them are values of the truncated model at that cutoff (status: *finite-N*). Nine cutoffs up to $N = 24$ are too few, and the behaviour varies too much with Δ_σ, for a controlled extrapolation, so we quote no continuum value in $d = 2$. Steps are taken between successive cutoffs, $\delta(N) = \Delta_\varepsilon(N) - \Delta_\varepsilon(N - 2)$.

- $\Delta_\sigma \ge 0.45$: the last step $\delta(24)$ is 2.54×10^{-6}, 1.28×10^{-6} and 1.57×10^{-7} at 0.45, 0.475 and 0.495, and the steps shrink by 18–22% per step.
- $0.25 \le \Delta_\sigma \le 0.4$: Δ_ε rises monotonically, with last steps 4.30×10^{-4}, 3.80×10^{-4}, 2.16×10^{-4} and 5.86×10^{-5}, which shrink by $1 - (\delta(24)/\delta(16))^{1/4} = 16$, 13, 11 and 8% per step. If the rise continues, $\Delta_\varepsilon(24)$ bounds the limit of the truncated model from below.
- $\Delta_\sigma = 0.2$: the steps change sign (+ up to $N = 22$, -4.2×10^{-5} at $N = 24$), so Δ_ε is not monotone.
- $0.125 \le \Delta_\sigma \le 0.175$: Δ_ε falls from $N = 10$ on, and the steps grow. The last step is -1.05×10^{-3} at 0.175 ($N = 24$) and -2.02×10^{-2} at 0.125 ($N = 22$), each larger than the previous one. At $\Delta_\sigma = 0.125$ the value at $N = 22$, 0.9400, lies 6% below the exact value 1 and moves away from it. This is the failure toward the short-range end discussed in Section 8.

5.5 No solution at $N = 24$ for $\Delta_\sigma = 0.125$, 0.126: a fold

At $\Delta_\sigma = 0.125$ and 0.126 the Newton solve at $N = 24$ did not converge. It was started from the $N = 22$ root and from the linear extrapolation $2\theta(22) - \theta(20)$, and repeated with a finite-difference Jacobian. Each time the residual stalled on a floor, at 2.9×10^{-3} and 2.0×10^{-3}. At the accepted roots of smaller N the condition number of the tuning Jacobian grows, from 84 at $N = 12$ to 888 at $N = 22$ ($\Delta_\sigma = 0.125$).

To decide whether a root exists, we scanned the valley of the residual. Write the conditions as

$$f_1(\theta) = E_\sigma - E_v - \Delta_\sigma, \qquad f_2(\theta) = E_{\phi^3} - E_v - (2 - \Delta_\sigma). \tag{20}$$

At six values $\theta_0 \in \{12.5, 13.25, \theta_0^{\text{best}}, 14.3, 14.8, 15.5\}$ around the best point of the Newton solves, a secant iteration in θ_1 solves $f_2 = 0$ to $|f_2| \le 10^{-9}$, and we record f_1 there. Code, basis and solver settings are those of production, with $n_z = 16$. At θ_0^{best} the first evaluation, before the secant, reproduces the stalled residual: $\max(|f_1|, |f_2|) = 2.93 \times 10^{-3}$ and 1.99×10^{-3}. A root would show as a sign change of f_1 along the curve $f_2 = 0$. There is none: $f_1 > 0$ at every point (smallest value 1.60×10^{-3}), with an interior minimum and a steep rise on both sides. A quartic fit in θ_0 puts the minimum at $f_1 = 2.686 \times 10^{-3}$ ($\theta_0 \approx 14.03$) for $\Delta_\sigma = 0.125$ and at 1.556×10^{-3} ($\theta_0 \approx 14.06$) for 0.126. Both are far above the effect of the zero-mode truncation ($\le 5 \times 10^{-7}$) and the eigensolver accuracy.

Hence, on the scanned range of θ_0, the two tuning conditions have no solution at $N = 24$ on the branch continued from smaller N. This is a statement about the truncated model at $N = 24$ and involves no extrapolation in N. The minimum falls by 1.13×10^{-3} per 0.001 in Δ_σ. Near a saddle-node (fold) it is linear in the parameter, which places the fold at

$$\Delta_\sigma^{\text{fold}}(N = 24) \approx 0.1274, \tag{21}$$

an estimate from two values of Δ_σ. Roots exist above it ($\Delta_\sigma = 0.135$ converged at $N = 24$), and at $N = 22$ they exist down to 0.125. The fold has therefore moved past 0.125 between $N = 22$ and 24, and the growing condition number at the roots is its precursor. The $d = 2$ data at $\Delta_\sigma = 0.125$ and 0.126 end at $N = 22$.

5.6 Weak coupling and the ϵ expansion of the model

Expanded in ϵ, the truncated model agrees with the consistent-window scheme of the perturbative calculation (`HT_Pert/`). At $\Delta_\sigma = 0.495$ ($\epsilon = 0.02$) the [4/4] Padé approximant of that scheme's coefficients $C_0(N), \ldots, C_8(N)$ reproduces $\Delta_\varepsilon(N)$ to 9.4×10^{-12} at every $N = 8, \ldots, 24$, and at $\Delta_\sigma = 0.475$ ($\epsilon = 0.1$) to 7.6×10^{-8} (check `d2.pert_consistent`). A difference δC_4 in the fourth-order coefficient would show as $\delta C_4\, \epsilon^4$, so $|\delta C_4| \lesssim 6 \times 10^{-5}$ at every $N \le 24$. In that scheme the fourth-order coefficient grows with the cutoff, $C_4(N) \simeq 5.003 - \frac{365}{4374\pi} \ln N$, that is like $-0.027 \ln N$ (exact sums to $N = 2000$ in `HT_Pert/`), while the continuum value $C_4(2)$ is finite. The $d = 2$ window model is thus known to be correct only through third order in ϵ. In the perturbative calculation a larger correction, the complete window of `HT_Pert/`, removes this logarithm; it has no finite-coupling counterpart here.

6 Three dimensions

6.1 Modes, sectors and the truncated space

On $S^2 \times \mathbb{R}$ of unit radius we use the modes $k = (n, l, m)$ of Eq. (2), with $n, l \ge 0$ and $|m| \le l$, level $\kappa = 2n + l$ and energy $\Delta_\sigma + \kappa$. With $z = e^{-\tau}$ and μ the cosine of the angle between the two points,

Δ_σ	s	$N=16$	$N=20$	$N=22$	$N=24$	last step	behaviour
0.125	1.75	0.98916082	0.96025057	0.94002832	n.s.	-2.02×10^{-2}	falls; steps grow
0.126	1.748	0.99053540	0.96363343	0.94568294	n.s.	-1.80×10^{-2}	falls; steps grow
0.135	1.73	0.99948557	0.98399079	0.97537120	0.96202900	-1.33×10^{-2}	falls; steps grow
0.15	1.7	1.00548133	0.99836315	0.99471461	0.99022622	-4.49×10^{-3}	falls; steps grow
0.175	1.65	1.00330217	1.00162583	1.00076308	0.99971675	-1.05×10^{-3}	falls; steps grow
0.2	1.6	0.99490861	0.99526003	0.99534664	0.99530455	-4.21×10^{-5}	not monotone
0.25	1.5	0.97541470	0.97674862	0.97727730	0.97770747	$+4.30\times10^{-4}$	rises; shrink 16%/step
0.3	1.4	0.96158682	0.96267180	0.96311194	0.96349241	$+3.80\times10^{-4}$	rises; shrink 13%/step
0.35	1.3	0.95679429	0.95736832	0.95760910	0.95782471	$+2.16\times10^{-4}$	rises; shrink 11%/step
0.4	1.2	0.96152920	0.96167002	0.96173269	0.96179129	$+5.86\times10^{-5}$	rises; shrink 8%/step
0.45	1.1	0.97526784	0.97527566	0.97527858	0.97528111	$+2.54\times10^{-6}$	small steps
0.475	1.05	0.98580651	0.98581110	0.98581268	0.98581396	$+1.28\times10^{-6}$	small steps
0.495	1.01	0.99678104	0.99678155	0.99678174	0.99678190	$+1.57\times10^{-7}$	small steps

Table 4: $\Delta_\varepsilon(N)$ in $d=2$ at finite coupling ($n_z=16$, consistent window): finite-N values of the truncated model, not continuum values. n.s.: no solution at $N=24$ (Section 5.5). Last step: between the two largest cutoffs. Shrink: $1-(\delta(24)/\delta(16))^{1/4}$.

the two-point function of the generalized free field is

$$\frac{z^{\Delta_\sigma}}{(1-2z\mu+z^2)^{\Delta_\sigma}}=\sum_{\kappa\geq0}z^{\Delta_\sigma+\kappa}\,G_\kappa(\mu),\qquad G_\kappa(\mu)=\sum_{2n+l=\kappa}(2l+1)\,u_{nl}^2\,P_l(\mu). \tag{22}$$

Here $G_\kappa=C_\kappa^{(\Delta_\sigma)}$ is the Gegenbauer polynomial, written G to avoid a clash with the coefficients C_n, and P_l is the Legendre polynomial. Expanding G_κ in Legendre polynomials fixes the weights u_{nl}^2 of Eq. (2). The zero mode $k=(0,0,0)$ has $u=1$; it is the oscillator a_0 of Eq. (6), with coordinate q. The bath field is $\chi=\sum_{k\neq(0,0,0)}u_{nl}\sqrt{4\pi}\,\big[Y_{lm}\,a_k+\text{h.c.}\big]$, and sphere averages use $d\Omega/4\pi$. A bath state with occupation numbers n_k has level $\ell=\sum_k\kappa_k n_k$. We keep all states with $\ell\leq N$, with no limit on the particle number.

Sectors. Rotations, inversion $(-1)^\ell$ and field parity commute with H. The scalar operators live in the $J=0$, inversion-even sector, in both field parities. Inversion-even states have even ℓ, so an odd cutoff adds no scalar state, and we use even N only. The $J=0$ subspace is the kernel of J_+ within the $J_z=0$ subspace. It is built separately in every species-occupation block, a species being (n,l) with its $2l+1$ magnetic states, and it is computed as an SVD null space.

At $N\geq18$ LAPACK's default SVD driver (gesdd) fails to converge on a few blocks. The code then retries with the gesvd driver and full right singular vectors. It accepts the kernel Q only if its dimension is the nullity predicted by representation theory and if every matrix element of J_+Q and of $Q^{\mathsf{T}}Q-1$ is at most 10^{-10} in absolute value; a second failure is an error. The retry happened in 8 of 17 335 blocks at $N=18$ and in 62 of 40 033 at $N=20$, with $\max|J_+Q|=5.8\times10^{-15}$ (it does not depend on Δ_σ). The scalar bath space has 110, 371, 1285, 4486, 15 815, 55 859 and 196 585 states at $N=8,10,\ldots,20$; with $n_z=16$ zero-mode states per parity, the scalar problem at $N=20$ has dimension 3.1×10^6 in each field parity.

Vertices. The couplings $W_j=\langle{:}\chi^j{:}\rangle$ of Eq. (7), restricted to $J=0$, equal the vertex at any single point of the sphere, by rotation invariance. The code evaluates them at the north pole. The $J=1$ and $J=2$ sectors are computed only as diagnostics; there the code uses the trace over the $2J+1$ magnetic components, and the original polar Gauss–Legendre quadrature (with $N+1$ nodes) is kept as an independent test. The vertices at lower cutoffs, which the window needs, are exact

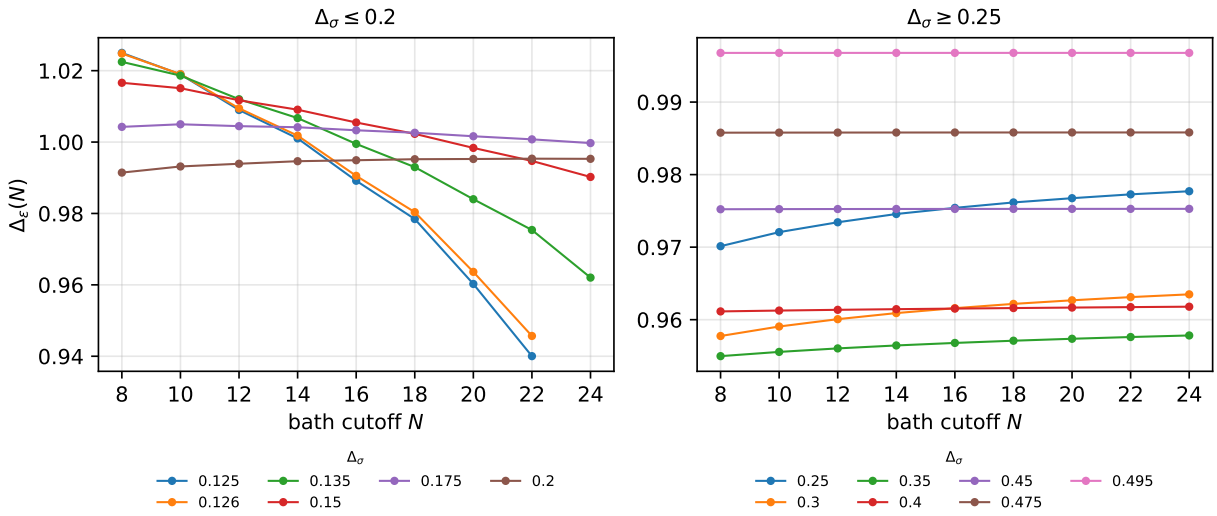


Figure 2: $\Delta_\varepsilon(N)$ in $d = 2$ against the bath cutoff N, for the 13 values of Δ_σ (made by `tools/make_d2_note_inputs.py` from `HT/results/HT_2d/prod1/points.csv`).

restrictions of those at N, because a normal-ordered W_j never passes through a level above both of its endpoints.

6.2 Window, zero-mode basis and tuning

The window of Eq. (11) is used in its self-consistent form. Only even sub-cutoffs carry distinct vacua, so a bath state of level ℓ is shifted by

$$D_N(\ell) = E_v(N) - E_v\big(2\lfloor (N-\ell)/2 \rfloor\big), \qquad E_v(M) = \lambda_{\min}\big[H(M) + D_M\big], \quad E_v(0) = 0, \tag{23}$$

with $E_v(M)$ the lowest field-even scalar energy at sub-cutoff M, solved one sub-cutoff after the other. At each sub-cutoff the fixed point $v = E_v(M)$ solves $F(v) = \lambda_{\min}[H(M) + D(v)] - v = 0$. Newton's method converges on it because $F'(v) = -p_0$, with p_0 the bath-vacuum weight of the ground state. The iteration stops when $|F| \le \min\big(10^{-12}\max(1,|v|),\, 10^{-11}\big)$ and fails if $p_0 \le 10^{-12}$. The same D_N is applied, without retuning, in the $J = 1, 2$ sectors.

The zero-mode basis is the density basis of Section 2.5, trained once per Δ_σ. The teacher is a solve at $N_t = 6$ in the energy basis, with 24 states per parity taken from an auxiliary oscillator basis of $N_{\rm osc} = 192$ states. The reduced density matrix of Eq. (10) is then $(\rho_{\rm vac} + \rho_\sigma + \rho_{\phi^3})/6$ at N_t plus the vacua at $M = 0, 2, 4$, each with weight $1/6$. The stored basis keeps its 20 leading eigenvectors per parity, and every production solve at $N = 8, \ldots, 20$ uses the leading $n_z = 16$. As in $d = 1$, the ε state plays no role in the training.

The couplings (r, g) are tuned by Eq. (4), $E_\sigma - E_v = \Delta_\sigma$ and $E_{\phi^3} - E_v = 3 - \Delta_\sigma$, by safeguarded Newton steps with tolerance 3×10^{-11}. Each cutoff is started from the solution at $N - 2$, with the same zero-mode basis. A point is accepted only if

- both tuning conditions hold to 1.5×10^{-10} (five times the Newton tolerance);
- the eigenvector residuals are at most 10^{-8} and their orthogonality defects at most 10^{-9};
- every window fixed point holds to 10^{-11}.

The replay also checks that the two tuned gaps of every record equal their targets to 10^{-9}. The prediction is $\Delta_\varepsilon = E_\varepsilon - E_v$, the first excited field-even scalar gap.

6.3 Campaigns

We solve 15 values of Δ_σ: 0.7499, 0.745, 0.74, 0.725, 0.7, 0.675, 0.65, 0.625, 0.6, 0.575, 0.55, 0.535, 0.525, 0.52 and 0.518149. The last lies 1.9×10^{-7} above the Δ_σ of the 3d Ising model [3], which is not imposed. Each value is computed as one sequence of solves, the teacher at $N = 6$ followed by $N = 8, 10, \ldots, 20$, each started from the previous one. Everything ran on SDSC Expanse (AMD EPYC 7742). Before production, each installed code release had to pass a validation job, the test suite and validators that `make test` runs. Every step then ran with the arguments fixed in advance in a campaign manifest (`code/lri_3d_window_hpc/public/PROVENANCE.md`).

The records are grouped in five campaigns, which name the folders of `data/HT_3d/campaigns`:

- `grid18` ($\Delta_\sigma \geq 0.52$, 14 values) and `grid18sri` ($\Delta_\sigma = 0.518149$, near the short-range Ising end): the teacher and $N = 8$ to 18. Each sequence used 16 cores. A point at $N = 18$, with $J = 1$ diagnostics, took 34–74 minutes at a peak of 38.5 GiB.

- `l20x06` ($\Delta_\sigma = 0.6$, with $J = 1$ diagnostics), `grid20` (the other 13 values $\Delta_\sigma \geq 0.52$) and `grid20sri` ($\Delta_\sigma = 0.518149$): $N = 20$, the last two in the scalar sector only. The bath graph and the scalar projector do not depend on Δ_σ, so they were built once into a cache that is never modified afterwards. On one large-memory node this took 9.0 h and 92 GiB, mostly in single-threaded LAPACK (from the Slurm log of a setup run, not a shipped record). The vertices depend on Δ_σ and are built in every point. A scalar-only point took 27–104 minutes at 84–87 GiB on 32 cores; the point at $\Delta_\sigma = 0.6$ with $J = 1$ diagnostics took 239 minutes at 328 GiB.

$N = 22$ was not attempted.

6.4 Results

Table 5 gives $\Delta_\varepsilon(N)$ at $N = 14$–20; all of $N = 8$–20 is in `results/HT_3d/finite_L_table.csv` (check `TABLE`). Figure 3 shows the steps $d_N = \Delta_\varepsilon(N) - \Delta_\varepsilon(N-2)$. We reach $N = 20$ at every Δ_σ and give finite-N values only. These cutoffs are too few for a controlled extrapolation, and the checks below show that small steps do not imply convergence.

- *Weak coupling, $\Delta_\sigma \geq 0.74$: small steps.* The last step is below 10^{-7} (7.3×10^{-8} at most).

- $\Delta_\sigma = 0.7$*: not converged, and moving away.* The Borel–Padé resummation of the ϵ series through the GFF coefficient $C_7(3)$ [4] fixes the continuum value to 1.47214(3). This is the median and half-spread of 29 approximants, with Borel–Leroy $b = 0, \ldots, 4$ and Padé orders of total degree 4 and 5, rejecting those with a pole on the positive axis. The value at $N = 20$, 1.4719248, lies 2.2×10^{-4} below it, and the sequence $\Delta_\varepsilon(16) > \Delta_\varepsilon(18) > \Delta_\varepsilon(20)$ moves away. The continuum is reached, if at all, only after a turnover at larger N. An extrapolation in N that continues this trend lands below $\Delta_\varepsilon(20)$, further from the continuum, so none is quoted.

- $\Delta_\sigma = 0.65$ *and* 0.675*: no trend yet.* The steps do not yet shrink: $d_{20}/d_{18} = 0.99$ and 1.23. At 0.7 and 0.725 the values fall (last steps -4.2×10^{-6} and -7.6×10^{-7}), and from 0.675 down they rise, so the steps change sign in this range.

Δ_σ	$N = 14$	$N = 16$	$N = 18$	$N = 20$	d_{20}	d_{20}/d_{18}
0.7499	1.49993336	1.49993336	1.49993336	1.49993336	2.3×10^{-11}	0.78
0.745	1.49673611	1.49673617	1.49673622	1.49673626	3.7×10^{-8}	0.75
0.74	1.49360198	1.49360214	1.49360225	1.49360232	7.3×10^{-8}	0.68
0.725	1.48486049	1.48485983	1.48485909	1.48485833	-7.6×10^{-7}	1.03
0.7	1.47194111	1.47193446	1.47192895	1.47192480	-4.2×10^{-6}	0.75
0.675	1.46046754	1.46048280	1.46050270	1.46052707	2.4×10^{-5}	1.23
0.65	1.45028710	1.45039728	1.45050697	1.45061574	1.1×10^{-4}	0.99
0.625	1.44146308	1.44172312	1.44196302	1.44218479	2.2×10^{-4}	0.92
0.6	1.43396549	1.43439430	1.43477460	1.43511242	3.4×10^{-4}	0.89
0.575	1.42767821	1.42827172	1.42878594	1.42922982	4.4×10^{-4}	0.86
0.55	1.42244491	1.42318494	1.42381704	1.42434921	5.3×10^{-4}	0.84
0.535	1.41973557	1.42054979	1.42124152	1.42181519	5.7×10^{-4}	0.83
0.525	1.41808151	1.41893773	1.41966339	1.42025885	6.0×10^{-4}	0.82
0.52	1.41729470	1.41816931	1.41890994	1.41951431	6.0×10^{-4}	0.82
0.518149	1.41700967	1.41789062	1.41863643	1.41924372	6.1×10^{-4}	0.81

Table 5: Δ_ε from the $d = 3$ window truncation at bath cutoffs $N = 14$–20, with the last step $d_{20} = \Delta_\varepsilon(20) - \Delta_\varepsilon(18)$ and its ratio to the previous one. Finite-cutoff values, not limits.

- $0.518 \le \Delta_\sigma \le 0.625$*: rising.* The values rise monotonically from $N = 8$ to 20. The last steps are $(2.2\text{–}6.1) \times 10^{-4}$, and they shrink by 8 to 19 percent per step. They shrink fastest near the short-range end.
- *Short-range end,* $\Delta_\sigma = 0.518149$*: fails near the endpoint.* The value at $N = 20$, 1.41924, lies above the 3d Ising value $\Delta_\varepsilon = 1.41263$ [3] and still rises. Near this end the long-range model is described by the short-range Ising model coupled, through its spin field, to an auxiliary generalized free field [8, 2]. That coupling is nearly marginal here, and the approach to the limit is expected to be logarithmic in N.
- *The window model at fourth order: a finite-N statement.* Expanded in ϵ at fixed N, the same window model has $C_2(N) \to C_2(3)$, and $C_3(N)$ is consistent with $C_3(3)$ (`HT_Pert/`). Its fourth-order coefficient at $N = 20$, 0.7400078, is 60% of $C_4(3) = 1.228063545$ [4] and still growing slowly with N; whether its $N \to \infty$ limit is wrong is not shown.

The $N = 20$ values are the $d = 3$ Hamiltonian-truncation results of Ref. [4]: rounded to five decimals in its $d = 3$ results table, and plotted as green squares in the right panel of its comparison figure.

7 Independent checks

In each dimension a second solver, written without using the production code, reproduces the production spectra at finite cutoff. These checks verify that we compute the stated model, not that its limit is the CFT.

One dimension.

- *Authorship.* A Claude Code subagent wrote the solver (`code/lri_dressed_phi4_independent_check`) from the definition of the model. It was not allowed to read the production code, results or analysis. Of the package documentation it read only the definition of the window (`docs/WINDOW.md`) and of the Hamiltonian (`docs/ALGORITHM.md`). It derived H itself from the GFF radial quantization, and its derivation agrees with Eqs. (8)–(9).

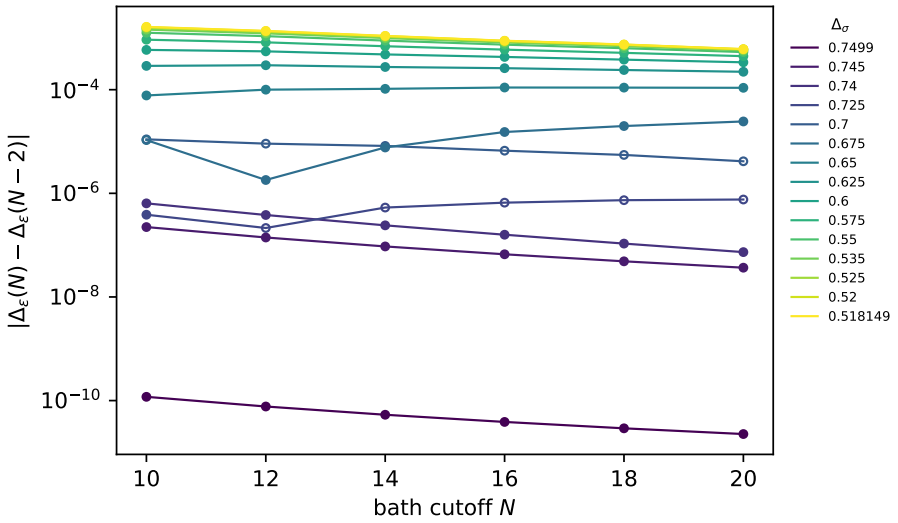


Figure 3: Size of the steps d_N against N, for every Δ_σ; filled symbols: $d_N > 0$, open: $d_N < 0$. Made by `public/note_d3.py` from `finite_L_table.csv`.

- *Own numerics.* The zero mode is a sinc discrete-variable representation, the bath Fock space is its own, and both the literal and consistent windows are implemented. Newton tuning continues from weak coupling. Every point is verified on 1.5 times finer and 1.5 times wider grids.
- *Comparison set.* 38 points: raw and consistent at $\Delta_\sigma = 0.249$, 0.2, 0.1, 0.05, 0.025 and 0.01 with $N = 12$, 16 and 20, plus literal at 0.2 and 0.1 with $N = 12$.
- *Agreement with the production records* (`dens_v1`, density basis, $n_z = 16$). At the 18 consistent-window points ($N = 12$, 16, 20) the records agree with the blind solver to 5.6×10^{-7} or better, the truncation error of the density basis.
- *Agreement with the energy-basis records* (`prod1--7`).
  - Δ_ε agrees to 1.3×10^{-9} where these records used $n_z = 48$ or $\Delta_\sigma \geq 0.2$.
  - It agrees to 4.8×10^{-9} where they used $n_z = 24$ or 32 at strong coupling, which is that basis's truncation error.
  - Over all 38 points this gives 5×10^{-9}. The other levels and the tuned (g, r) agree at the same level.
- *Weak coupling.* The solver reproduces $C_2(N)$, Eq. (16), to 3×10^{-7}. It also shows that the window changes Δ_ε only at order ϵ^4: the fitted exponent is 4.05 to 4.11.

Two dimensions. A second solver, `lri_2d_window_indep`, implements the $d = 2$ model of Section 5.1 from its written definition, without using the production code. Its numerics differ:

- bath vertices from explicit annihilation amplitudes, on a dictionary representation of the Fock states;
- reflection-even states as explicit symmetric combinations;

- the zero mode in the energy basis, the lowest eigenstates of H_z with K states per parity, instead of the density basis;
- its own Newton iteration for the window fixed points, and tuning by a finite-difference Newton method.

It is not blind: its author had earlier prepared the production code for the cluster. Its self-tests compare the vertices with a brute-force construction in the full Fock space at $N = 5$, and its Lanczos solver with dense diagonalization. At the production couplings it reproduces the production spectra (155 runs at 13 values of Δ_σ; each comparison is bound to the shipped record by its hash):

- With a converged zero-mode basis ($K \geq 32$) it reproduces Δ_ε to 1.4×10^{-12} at $N = 12$ and 16 for every Δ_σ, and the five lowest even and odd levels and the whole vacuum table $E_v(M)$ to 7.6×10^{-12}. At $N = 18$ ($\Delta_\sigma \leq 0.15$) Δ_ε agrees to 4.5×10^{-12}, and at $N = 20$ ($\Delta_\sigma \leq 0.175$) to 4.6×10^{-11}.
- At $N = 22$, $\Delta_\sigma = 0.125$ and 0.126, it agrees to 4.4×10^{-10} and 4.0×10^{-10} ($K = 32$). There the differences grow by a factor of about nine from one cutoff to the next and keep one sign. This is the signature of the truncation at $K = 32$, not of either code.
- With $K = 24$ the agreement degrades at strong coupling as K runs out (6×10^{-9} at $N = 16$ and 2.4×10^{-7} at $N = 20$, $\Delta_\sigma = 0.125$). This tests the check rather than production.
- Re-tuned independently from couplings shifted by $(0.05, -0.05)$ in θ, the solver returns to the production root: Δ_ε agrees to 9×10^{-11} at $N \leq 12$, and to 3.6×10^{-8} at $N = 16$ with $K = 24$.

The production zero-mode basis is itself converged: $n_z = 20$ and 16 agree to 9.4×10^{-11} over all chains and cutoffs. These checks verify that we compute the stated model; they do not test its continuum limit.

Three dimensions. A second solver, written from the definition of the model without reading or importing the production code (`code/lri_3d_window_indep`), reproduces the $d = 3$ spectra. It makes different choices at every step:

- a complex Y_{lm} Fock space with all magnetic sectors;
- the $J = 0$ space as the null space of the Casimir $J_- J_+$, from a dense eigensolver, rather than an SVD null space;
- the vertices at a generic point of the sphere, checked at a second point;
- the zero mode in a number-state basis truncated to its K lowest local eigenstates, not the trained density basis;
- window fixed points by bracketed root finding.

Its $J = 0$ state counts at $N = 2, 4, 6, 8$ (3, 10, 33, 110) equal those of the perturbative package in `HT_Pert/`. In 23 comparisons at $N = 8, 10, 12$, in the window and the raw scheme and over the whole range of Δ_σ, it reproduces the production Δ_ε:

- at the production couplings to at most 1.8×10^{-12}, with 16 or 24 local states;
- after tuning (r, g) independently from a start displaced by 2%, to at most 1.9×10^{-11}, quoted as 2×10^{-11}, with every tuning converged to 10^{-12}.

The production values it compares with are those of the shipped records. This verifies that the stated model is computed at $N \leq 12$, not that its limit is the CFT. The quoted results are at $N = 14$–20, beyond the reach of this check.

8 Caveats

The limit depends on the window model.

- The window restores only disconnected vacuum fluctuations.
- In $d = 1$ the corrected descendant spacings still miss by amounts that do not decrease with N (Table 3).
- In $d = 2$ the ϵ expansion of the window model is correct only through third order: its fourth-order coefficient grows like $\ln N$ (Section 5). In $d = 3$ the fourth-order coefficient at $N = 20$ is 60% of $C_4(3)$ and still growing slowly; whether its $N \to \infty$ limit is wrong is not shown (Section 6).

The method fails toward the short-range end in every dimension.

- The tuned coupling is largest there.
- In $d = 1$ it also grows with the cutoff, and toward the short-range end $\Delta_\sigma \to 0$ the σ gap that the tuning imposes, $E_\sigma - E_v = \Delta_\sigma$, closes.
- The failure sets in at $\Delta_\sigma \leq 0.05$ in $d = 1$, at $\Delta_\sigma \leq 0.175$ in $d = 2$, and near the endpoint in $d = 3$.

Errors.

- The quoted errors are half-spreads of the extrapolations in the cutoff, not standard deviations, and they do not include the dependence on the window prescription (Section 4.4).
- The exponents of Eq. (14) are fitted, not derived.
- At $\Delta_\sigma \geq 0.15$ in $d = 1$ the spreads are probably optimistic.
- The classification *final* versus *one-sided* is a judgement based on the local exponent.

Order of the analysis.

- The window was designed after its effect had been seen in the descendant spacings.
- The fit protocol of Section 4.2 was carried over unchanged from the earlier `prod1--7` analysis.
- The held-out test and the local exponent were added after the $N = 40$ data had been seen.
- The tail-model and prescription comparisons of Section 4.4 were made after the gap to NRG had been seen. For that reason they are given as shifts and are not turned into an error.

One dimension.

- *Strong coupling.* The $d = 1$ values at $\Delta_\sigma \leq 0.05$ are not limits. At 0.025 and below, the correction overshoots: the sequence turns over and falls without bound in $\ln N$.

- *Comparison with NRG.* Where the descriptions overlap ($0.075 \le \Delta_\sigma \le 0.2$), HT lies above NRG at all six points, by up to 1.1×10^{-3}. At $\Delta_\sigma = 0.1$ and 0.125 the gap exceeds the spread of our extrapolations. However, the uncorrected prescription extrapolates to limits 7×10^{-4}–2.3×10^{-3} lower, below NRG, so the dependence on the prescription is as large as the gap (Section 4.4).

- *Prescription dependence.* The quoted errors of Table 1 do not include the prescription dependence. The raw and window limits differ by $6.6 \times 10^{-4}, 2.3 \times 10^{-3}, 1.7 \times 10^{-3}$ at $\Delta_\sigma = 0.125$, 0.1 and 0.075.

Two dimensions. All $d = 2$ values are finite-N values of the truncated model. With cutoffs up to $N = 24$, and a behaviour that changes with Δ_σ, we quote no continuum limit and no extrapolation error. The window model is known to be correct only through third order in ϵ: expanded in ϵ, the model agrees with the consistent-window scheme, whose fourth-order coefficient grows like $-0.027 \ln N$ (Section 5.6). Even at weak coupling, the $N \to \infty$ limit of the truncated model is therefore not the continuum beyond order ϵ^3. Toward the short-range end the method fails: at $\Delta_\sigma \le 0.175$ the values fall with growing steps, at $\Delta_\sigma = 0.125$ the value at $N = 22$ lies 6% below the exact $\Delta_\varepsilon = 1$ and moves away, and at $N = 24$ the tuning conditions have no solution for $\Delta_\sigma \lesssim 0.1274$. The rise at $0.25 \le \Delta_\sigma \le 0.4$ bounds the truncated model's limit from below only if it continues.

The fold location is an estimate from two values of Δ_σ, and it assumes a saddle-node. The absence of a root at $N = 24$ is established on the scanned range $12.5 \le \theta_0 \le 15.5$, where f_1 has a single interior minimum and rises steeply at both ends. A root on another branch, far from the one continued from smaller N, is not excluded. Finally, the independent solver is not blind (Section 7).

Three dimensions.

- We reach $N = 20$ at every Δ_σ, too few cutoffs for a controlled extrapolation, and quote finite-N values only. Small steps do not mean convergence. At $\Delta_\sigma = 0.7$ the sequence moves away from the resummed ϵ series; it reaches the continuum, if at all, only after a turnover at larger N, so extrapolations of these data are biased low. At 0.65 and 0.675 the steps do not yet shrink.

- Expanded in ϵ, the window model has $C_2(N)$ and $C_3(N)$ approaching their continuum values (`HT_Pert/`). Its fourth-order coefficient at $N = 20$ is 60% of $C_4(3)$ and still growing slowly with N; whether its $N \to \infty$ limit is wrong is not shown.

- The method fails toward the short-range end, where the tuned couplings are largest. At $\Delta_\sigma = 0.518149$ the $N = 20$ value lies 6.6×10^{-3} above the 3d Ising value [3] and still rises; there an approach logarithmic in N is expected.

- The cost limits the cutoff. At $N = 20$ the scalar SO(3) projector takes about 9 h of single-threaded LAPACK, and an $N = 20$ point needs 84–87 GiB, or 328 GiB with the $J = 1$ diagnostics. $N = 22$ was not attempted.

- Four changes to the Expanse campaign tooling of `lri_3d_window_hpc` were not independently inspected: the fixes of the second Codex inspection round (630a217) with a test-fixture fix (e895443), and two campaign options (ccd055a, b330ee7). They change only the cluster tooling and its tests. The numerical source is unchanged, and every record carries its hash (check `REC-SRC`). The production results had no separate review; the independent solver and `HT_Pert/` cross-check them (`code/lri_3d_window_hpc/public/REVIEWS.md`).

9 Reproduction

9.1 Layout

All paths are relative to the folder `HT/` of the public repository.

Folders.

- `code/<package>/`: one package per solver or analysis, shipped at the commit its production ran. Each has a `public.mk` with the targets `test` and `replay` (optionally `smoke`), and a `public/` folder with the replay scripts and a README.
- `data/HT_<d>d/`: the production records.
- `results/HT_<d>d/`: the published tables and the expected values of every quoted number.

Commands.

- `make test` and `make replay` in `HT/` run the targets of every package.
- In a package folder: `make -f public.mk test|replay|smoke`.

A replay recomputes every quoted number of its package from the shipped records, without a production rerun. It prints a PASS/FAIL table and exits non-zero on any failure.

9.2 Packages and records

package (`code/`)	d, role	content, records, cost
`lri_dressed_phi4_hpc`	1: solver	commit 33998bb. Zero mode plus bath, window, density basis, launcher for the symmetry cluster. Records `data/HT_1d/dens_v1` (201) and `prod_window` (680). `test` about 3 min; `smoke` one item
`lri_dressed_phi4_hpc_analysis`	1: analysis	fits, held-out test, identity records, tables. `replay` about 1 min
`lri_dressed_phi4_independent_check`	1: blind check	DVR solver, its per-task outputs and comparison. `replay` seconds
`lri_2d_window_hpc`, `lri_2d_window_indep`	2	Section 5; records `data/HT_2d`
`lri_3d_window_hpc`, `lri_3d_window_indep`	3	Section 6; records `data/HT_3d`

9.3 Rules that are part of the results

- *Estimators.* The fit families, acceptance and stability rules, and estimators of Section 4.2 define the quoted numbers. So do the status words of Table 1.
- *Expected values.* `results/HT_1d/quoted_numbers.json` gives the estimator and comparison rule of every $d = 1$ number. The $d = 1$ numbers in this note are typeset from it.
- *Fit tolerance.* The cutoff fits of $d = 1$ are nonlinear least squares from several starting exponents, and other numpy/scipy versions may land in slightly different minima. The replay therefore checks each number twice:

– against the shipped analysis outputs, exactly;
– against a refit in the reader's environment, within a quarter of the quoted half-spread.

The largest shift measured on symmetry was 0.11 of the half-spread ($\Delta_\sigma = 0.175$).

9.4 Result → code, record, command, tolerance

result	value	pkg	records	command	tolerance
d = 1, *finite coupling* (A = `lri_dressed_phi4_hpc_analysis`, I = `lri_dressed_phi4_independent_check`; records D = `data/HT_1d/dens_v1`, R = `data/HT_1d/prod_window`, B = the stored bases in D, ID = `identities/` in A, S = the solver `lri_dressed_phi4_hpc`, NRG = `results/HT_1d/nrg_reference.json`; expected values `results/HT_1d/quoted_numbers.json`; tolerance "status: the stability flag behind the status word)					
smallest cutoff of the fitted series	12	A	D	replay `S1`	exact
largest cutoff, all 13 Δ_σ	40	A	D	replay `S2`	exact
$N_{\min}$ of the fits (smallest, largest)	12–24	A	D	replay `S3`	exact
fits per Δ_σ (F1, F2, F3 × 7 $N_{\min}$)	21	A	D	replay `S4`	exact
density-basis states per parity in production	16	A	D	replay `S5`	exact
training cutoff of the density basis	24	A	B	replay `S6`	exact
zero-mode states per parity of the teacher	48	A	B	replay `S7`	exact
states kept in each stored trained basis (dens16 = its first 16)	24	A	B	replay `S8`	exact
dens16 vs the largest-n_z reference (n_z 48/56) at N = 32: max |difference| ≤ bound	10^{-6}	A	D, R	replay `S9`	bound
stability rule: ≥ 5 fits, median $\omega \geq 0.45$, half-spread ≤ 2e-3 (dens_v1.py stable())	0.45, 2e-3	A	—	replay `S10`	judgement
median ω of the stable fits (min, max over Δ_σ)	0.6–0.9	A	D	replay `S11`	as quoted
raw level-1 violations, $\Delta_\sigma \leq 0.1$, N = 16..28, unambiguous (min, max)	0.06–0.25	A	ID	replay `V1`	as quoted
σ: |raw / window| level-1 violation at N = 28, $\Delta_\sigma \leq 0.15$ (min, max)	10–40	A	ID	replay `V2`	as quoted
window residual of σ and shadow, $\Delta_\sigma \leq 0.1$, N = 28: median $\|v_1\|$ ('about 0.01')	0.01	A	ID	replay `V3`	factor 2
window residual of ε, $\Delta_\sigma \leq 0.05$, N = 28: largest $\|v_1\|$ ('up to')	0.05	A	ID	replay `V4`	sig. figures
cutoffs of the raw identity records behind V1 ($\Delta_\sigma \leq 0.1$; min, max N)	16–28	A	ID	replay `V5`	as quoted
cutoff of the identity table (Table 3, copied in Ref. [4]) and of V2–V4: largest N of the identity records	28	A	ID	replay `V6`	as quoted

result	value	pkg	records	command	tolerance
rows of the identity table at N = 28, Δ_σ = 0.2, 0.15, 0.1, 0.075, 0.05, 0.025: raw and window level-1 violations of σ, ϕ^3 and E, as printed	6 table rows (Δ_σ = 0.2 to 0.025)	A	ID	replay `V7`	exact
ω_{eff} at N $\sim$ 35, Δ_σ = 0.125 (levels off)	0.64	A	D	replay `E1`	as quoted
ω_{eff} at N $\sim$ 35, Δ_σ = 0.1 (levels off)	0.68	A	D	replay `E2`	as quoted
ω_{eff} at N $\sim$ 15 and $\sim$ 35, Δ_σ = 0.075 (rising)	0.70–0.80	A	D	replay `E3`	as quoted
ω_{eff} at N $\sim$ 15 and $\sim$ 35, Δ_σ = 0.05 (rising)	0.92–1.16	A	D	replay `E4`	as quoted
fits on N $\leq$ 32 predict N = 34..40: max |actual - median prediction|, $\Delta_\sigma \geq 0.05$	2.3×10^{-6}	A	D	replay `E5`	sig. figures
Δ_σ = 0.075: slope at N = 40 $\times$ (1/w - 1/(w + 0.2)), w = $\omega_{\text{eff}}(35)$	5×10^{-4}	A	D	replay `E6`	sig. figures
Δ_∞ - Δ_ε(N = 40) at Δ_σ = 0.075..0.125 (min, max)	1.6×10^{-3}–2.5×10^{-3}	A	D	replay `E7`	as quoted
Δ_σ = 0.025: cutoff of the largest Δ_ε(N)	22	A	D	replay `E8`	exact
Δ_σ = 0.025: d Δ_ε / d ln N between N = 38 and 40	-1.3×10^{-3}	A	D	replay `E9`	sig. figures
Δ_σ = 0.01: d Δ_ε / d ln N between N = 38 and 40 (constant)	-6.2×10^{-3}	A	D	replay `E10`	sig. figures
Δ_σ = 0.249: Δ_∞ - ϵ series through ϵ^7	-4×10^{-8}	A	D	replay `P1`	sig. figures
Δ_σ = 0.24: band of the partial sums through ϵ^4..ϵ^7	0.49419–0.49426	A	D	replay `P2`	as quoted
Δ_σ = 0.24: Δ_∞ lies inside that band	yes	A	D	replay `P3`	true
level at which the ϵ-series check is trusted	10^{-6}	A	—	replay `P4`	judgement
Δ_ε(N = 40), Δ_σ = 0.249	0.4993430	A	D	replay `T01`	as quoted
Δ_ε(N = 40), Δ_σ = 0.24	0.494186	A	D	replay `T02`	as quoted
Δ_ε(N = 40), Δ_σ = 0.225	0.487982	A	D	replay `T03`	as quoted
Δ_ε(N = 40), Δ_σ = 0.2	0.482839	A	D	replay `T04`	as quoted
Δ_ε(N = 40), Δ_σ = 0.175	0.483541	A	D	replay `T05`	as quoted
Δ_ε(N = 40), Δ_σ = 0.15	0.490388	A	D	replay `T06`	as quoted
Δ_ε(N = 40), Δ_σ = 0.125	0.504561	A	D	replay `T07`	as quoted
Δ_ε(N = 40), Δ_σ = 0.1	0.528210	A	D	replay `T08`	as quoted
Δ_ε(N = 40), Δ_σ = 0.075	0.564851	A	D	replay `T09`	as quoted
Δ_ε(N = 40), Δ_σ = 0.05	0.620671	A	D	replay `T10`	as quoted
Δ_ε(N = 40), Δ_σ = 0.025	0.710577	A	D	replay `T11`	as quoted
Δ_ε(N = 40), Δ_σ = 0.01	0.809111	A	D	replay `T12`	as quoted
Δ_ε(N = 40), Δ_σ = 0.001	0.969338	A	D	replay `T13`	as quoted
Δ_∞ (median of accepted fits, half-spread), Δ_σ = 0.249	0.49934358(9)	A	D	replay `D01`	as quoted
Δ_∞, Δ_σ = 0.24	0.4942198(26)	A	D	replay `D02`	as quoted
Δ_∞, Δ_σ = 0.225	0.488080(5)	A	D	replay `D03`	as quoted
Δ_∞, Δ_σ = 0.2	0.483024(5)	A	D	replay `D04`	as quoted

result	value	pkg	records	command	tolerance
Δ_∞, $\Delta_\sigma = 0.175$	0.483909(15)	A	D	replay `D05`	as quoted
Δ_∞, $\Delta_\sigma = 0.15$	0.491209(33)	A	D	replay `D06`	as quoted
Δ_∞, $\Delta_\sigma = 0.125$	0.50612(4)	A	D	replay `D07`	as quoted
Δ_∞, $\Delta_\sigma = 0.1$	0.53048(12)	A	D	replay `D08`	as quoted
Δ_∞, $\Delta_\sigma = 0.075$: half-spread, then the extra one-sided shift -5e-4 (E6) as a second bracket	0.56733(20)(−50)	A	D	replay `D09`	as quoted
$\Delta_\sigma = 0.05$: accepted F1 and F2 fits (min, max)	0.6220–0.6224	A	D	replay `D10`	as quoted
$\Delta_\sigma = 0.05$: lowest accepted F3 fit	0.618	A	D	replay `D11`	as quoted
total-error estimate at $\Delta_\sigma \geq 0.15$ (the fit spreads are optimistic)	10^{-4}	A	—	replay `D12`	judgement
largest fit half-spread at $\Delta_\sigma \geq 0.15$ ('at most' in the caption of the note's Table 1)	3.3×10^{-5}	A	D	replay `D13`	sig. figures
final vs one-sided ($\Delta_\sigma = 0.075$): judgement from the local exponent, still rising at 0.075 (E3: +0.10 over N $\sim$ 15..35) but levelling off at 0.125 and 0.1 (E1, E2); the computed rows U01–U13 check stable / unstable / no limit	one-sided at 0.075 only	A	—	replay `U00`	judgement
status, $\Delta_\sigma = 0.249$	final	A	D	replay `U01`	status
status, $\Delta_\sigma = 0.24$	final	A	D	replay `U02`	status
status, $\Delta_\sigma = 0.225$	final	A	D	replay `U03`	status
status, $\Delta_\sigma = 0.2$	final	A	D	replay `U04`	status
status, $\Delta_\sigma = 0.175$	final	A	D	replay `U05`	status
status, $\Delta_\sigma = 0.15$	final	A	D	replay `U06`	status
status, $\Delta_\sigma = 0.125$	final	A	D	replay `U07`	status
status, $\Delta_\sigma = 0.1$	final	A	D	replay `U08`	status
status, $\Delta_\sigma = 0.075$	one-sided	A	D	replay `U09`	status
status, $\Delta_\sigma = 0.05$	unstable	A	D	replay `U10`	status
status, $\Delta_\sigma = 0.025$	no limit	A	D	replay `U11`	status
status, $\Delta_\sigma = 0.01$	no limit	A	D	replay `U12`	status
status, $\Delta_\sigma = 0.001$	no limit	A	D	replay `U13`	status
max $\lvert\Delta_\infty$ - NRG$\rvert$ at $0.075 \leq \Delta_\sigma \leq 0.2$ (NRG: re-analysed final table, `results/HT_1d/nrg_reference.json`); at $\Delta_\sigma = 0.075$	1.1×10^{-3}	A	D, NRG	replay `C1`	sig. figures
ω_{eff} at N $\sim$ 15 and $\sim$ 35, $\Delta_\sigma = 0.249$ (slow rise at weak coupling)	0.39–0.46	A	D	replay `E11`	as quoted
$\Delta_\sigma = 0.249$: bias of a further rise of ω_{eff} by 0.2 (as E6) is below	10^{-5}	A	D	replay `E12`	bound
Δ_∞ - NRG at $\Delta_\sigma = 0.125$, 0.1, 0.075 (2 significant figures)	4.5×10^{-4}, 8.7×10^{-4}, 1.1×10^{-3}	A	D, NRG	replay `C2`	sig. figures

result	value	pkg	records	command	tolerance
$\Delta_\sigma = 0.001$: Δ_ε(N = 40) - NRG	0.032	A	D, NRG	replay `C3`	as quoted
A = $\Delta_\sigma/4$ + r - g/2 at the tuned couplings (dens16): largest Δ_σ at which its sign changes with the cutoff (A < 0 at every N for $\Delta_\sigma = 0.2$, 0.175; > 0 at $\Delta_\sigma \geq 0.225$)	0.15	A	D	replay `S12`	as quoted
energy zero-mode basis at Δ_σ = 0.025, N = 28, consistent window: Δ_ε for n_z = 32, 40, 48 (prod records)	0.710632, 0.710934, 0.710959	A	R	replay `B1`	as quoted
largest fit half-spread among the final results ($\Delta_\sigma \geq 0.1$)	1.2×10^{-4}	A	D	replay `D14`	sig. figures
largest tail-model shift (F4 with two free exponents; drifting local exponent) of the window limit at $0.075 \leq \Delta_\sigma \leq 0.125$	1.6×10^{-4}	A	D	replay `M3`	sig. figures
NRG lies in the bracket [$\Delta_\varepsilon(40)$, $\Delta_\varepsilon(40)$ + slope(40)/$\omega_{\text{eff}}(35)$] of the window series at every $0.075 \leq \Delta_\sigma \leq 0.2$	yes	A	D, NRG	replay `M4`	true
the raw limit lies below NRG and the window limit above it at $0.075 \leq \Delta_\sigma \leq 0.125$	yes	A	D, R, NRG	replay `M5`	true
window minus raw limit at Δ_σ = 0.125, 0.1, 0.075 (2 significant figures)	6.6×10^{-4}, 2.3×10^{-3}, 1.7×10^{-3}	A	D, R	replay `M6`	sig. figures
largest |window - raw| limit difference at Δ_σ = 0.2, 0.175, 0.15	1.3×10^{-4}	A	D, R	replay `M7`	sig. figures
smallest Δ_∞ - NRG at $0.075 \leq \Delta_\sigma \leq 0.15$ (at 0.15)	10^{-4}	A	D, NRG	replay `C5`	sig. figures
Δ_∞ - NRG at Δ_σ = 0.175, 0.2 is below 1e-4	10^{-4}	A	D, NRG	replay `C6`	bound
Δ_∞ > NRG at every $0.075 \leq \Delta_\sigma \leq 0.2$	yes	A	D, NRG	replay `C7`	true
Δ_∞ - NRG exceeds the fit half-spread at Δ_σ = 0.1 and 0.125	yes	A	D, NRG	replay `C8`	true
max |Δ_∞ - NRG| / NRG at $0.075 \leq \Delta_\sigma \leq 0.2$ is at most 2e-3	2×10^{-3}	A	D	replay `C9`	bound
window minus raw limit at $0.075 \leq \Delta_\sigma \leq 0.125$ (min with 1, max with 2 significant figures)	7×10^{-4}–2.3×10^{-3}	A	D, R	replay `M10`	sig. figures
blind solver vs production, 38 points	5×10^{-9}	I	I `results/`, D, R	replay `B6` (B4, B5)	1 sig. fig.
blind solver vs dens16, $N \leq 20$	$\leq 10^{-6}$ (5.6×10^{-7})	I	same	replay `B7`	bound
weak coupling $C_2(N)$, $N \leq 16$	$\leq 3 \times 10^{-7}$	I	I `results/`	replay `B8`	bound

result	value	pkg	records	command	tolerance
window at $O(\epsilon^4)$ in the blind solver	exponent ≈ 4	I	I `results/`	replay `B9`	3.8–4.3
records intact and bound to the code	1106 records	A, S	D, R	replay `R1`–`R7`, S `F1`, `F2`	exact
rows, refits, tables	as shipped	A	D	replay `W1`, `A0`–`A4`, `T1`	exact; refit $\leq \frac{1}{4}$ half-spread

$d = 2$, *finite coupling.* Packages: P = `lri_2d_window_hpc`, I = `lri_2d_window_indep`, HP = `HT_Pert/code/lri_2d_coeff_independent_check`. Records: R = `data/HT_2d/prod1/chains`, V = `results/HT_2d/valley_N24`, O = `results/HT_2d/independent_check/out_sym`, C = `data/HT_2d/pert_crosscheck/consistent_C0_C8.csv`. Command: replay = `make -f public.mk replay` in the package, followed by the check name.

record integrity; source hash = code 889a968	348 records	P	R	replay `d2.records.*`	exact
largest cutoff (22 at $\Delta_\sigma = 0.125$, 0.126)	$N = 24$	P	R	replay `d2.Nmax`	exact
$\Delta_\varepsilon(N)$, $N = 8, \ldots, 24$, Table 4, Fig. 2	13 chains	P	R; `results/HT_2d/prod1/points.csv`	replay `d2.points`	exact
values at the largest N (Ref. [4], Fig. `fig:cmp2d3d`, Table `tab:num2`)	13 values	P	`results/HT_2d/prod1/largest_N.csv`	replay `d2.largest_N`; `d2.num2`	exact; 5 decimals
last step, $\Delta_\sigma \geq 0.45$ (largest at 0.45)	$\leq 2.6 \times 10^{-6}$ (2.54×10^{-6})	P	R	replay `d2.conv_step`	bound
shrink per step, $\Delta_\sigma \geq 0.45$	18–22%	P	R	replay `d2.conv_shrink`	integer %
rise, $0.25 \leq \Delta_\sigma \leq 0.4$	monotone	P	R	replay `d2.mono.rises`	sign
last steps, $0.25 \leq \Delta_\sigma \leq 0.4$	$(0.6\text{–}4.3) \times 10^{-4}$	P	R	replay `d2.mono_steps`	2 digits
shrink per step, $0.25 \leq \Delta_\sigma \leq 0.4$	8–16%	P	R	replay `d2.mono_shrink`	integer %
$\Delta_\sigma = 0.2$	not monotone	P	R	replay `d2.nonmono_0.2`	sign
fall, $0.125 \leq \Delta_\sigma \leq 0.175$	falls, steps grow	P	R	replay `d2.fall`	sign, ratio
last step, $\Delta_\sigma = 0.175$	-1.0×10^{-3}	P	R	replay `d2.fall_small`	2 digits
last step, $\Delta_\sigma = 0.125$, $N = 20 \to 22$	-2.0×10^{-2}	P	R	replay `d2.fall_large`	2 digits
n_z convergence: $n_z = 20$; 12 against 16	9.4×10^{-11}; 5.2×10^{-7}	P	R	replay `d2.nz.*`	10^{-10}; 6×10^{-7}
reference points ($N = 14$) against $n_z = 16$	$\leq 10^{-9}$	P	R (`ref*`)	replay `d2.refs`	10^{-9}
Jacobian condition at the roots, $\Delta_\sigma = 0.125$, $N = 12$; 22	84; 888	P	R	replay `d2.cond`	integer
stalled Newton residual, $N = 24$, $\Delta_\sigma = 0.125$; 0.126	2.93×10^{-3}; 1.99×10^{-3}	P	V	replay `d2.valley.best`	2 digits
no root at $N = 24$, $\Delta_\sigma = 0.125$, 0.126: $f_1 > 0$ on $f_2 = 0$	smallest f_1: 1.60×10^{-3}	P	V	replay `d2.no_root`	sign
valley minima, $\Delta_\sigma = 0.125$; 0.126	2.686×10^{-3}; 1.556×10^{-3}	P	V	replay `d2.valley.min`	10^{-6}
fold	$\Delta_\sigma \approx 0.1274$	P	V	replay `d2.fold`	5×10^{-5}

result	value	pkg	records	command	tolerance
weak coupling: model agrees with the consistent scheme, $\Delta_\sigma = 0.495$; 0.475	9.4×10^{-12}; 7.6×10^{-8}	P	R, C	replay `d2.pert_consistent.*`	10^{-11}; 10^{-7}
$\ln N$ coefficient of $C_4(N)$, $-365/(4374\pi)$	-0.027	HP	exact $C_4(N)$, $N \le 2000$	replay `p2.c4_log_consistent`	set in HP
independent solver, $N = 12$, 16	1.4×10^{-12}	I	O, R	replay `i2.dE.N12_16`	2 digits
independent solver, levels	7.6×10^{-12}	I	O, R	replay `i2.levels.N12_16`	10^{-11}
independent solver, $N = 18$; 20	4.5×10^{-12}; 4.6×10^{-11}	I	O, R	replay `i2.dE.N18`; `N20`	5×10^{-12}; 5×10^{-11}
independent solver, $N = 22$, $\Delta_\sigma = 0.125$; 0.126	4.4×10^{-10}; 4.0×10^{-10}	I	O, R	replay `i2.dE.N22`	2 digits
independent re-tuning, $N \le 12$; $N = 16$	9×10^{-11}; 3.6×10^{-8}	I	O, R	replay `i2.retune.*`	10^{-10}; 4×10^{-8}
$d = 3$, *finite coupling* (P = `lri_3d_window_hpc`, I = `lri_3d_window_indep`, C = `data/HT_3d/campaigns`)					
records intact (`REC-*`)	105 records, 1 source hash	P	every `window_L*.json` in C; `data/HT_3d/MANIFEST.sha256`	`make replay`	sha256, self-hash, seed and basis chain, acceptance gates: exact
$\Delta_\varepsilon(N)$, $N = 8$–20, 15 Δ_σ (Table 5; comparison figure of Ref. [4]) (`TABLE`)	`finite_L_table.csv`	P	the 105 window records	`make replay`	byte-identical rebuild
largest cutoff (`NMAX`)	$N = 20$	P	15 `window_L20.json`	`make replay`	exact
$N = 20$ values in the $d = 3$ results table of Ref. [4] (`NUM3`)	15 values, 1.41924 … 1.49993	P	15 `window_L20.json`	`make replay`	5 decimals
setup: teacher $N_t = 6$, 24 states, $N_{\mathrm{osc}} = 192$, 20 stored; $n_z = 16$ (`CONFIG`)	as stated	P	`teacher_L6.json`, window records	`make replay`	exact
scalar bath states, $N = 8$–20 (`SIZE`)	110 … 196 585	P	window records (dimension/$2n_z$)	`make replay`	exact
projector retries (`PROJECTOR`)	$N = 18$: 8/17 335; $N = 20$: 62/40 033; $\lvert J_+Q\rvert < 10^{-14}$	P	$N = 18$, 20 window records	`make replay`	exact counts
cost at $N = 18$ (`COST-L18`)	34–74 min, 38.5 GiB	P	$N = 18$ window records	`make replay`	integer minutes, 0.1 GiB
cost at $N = 20$ (`COST-L20`)	27–104 min, 84–87 GiB; 239 min, 328 GiB	P	$N = 20$ window records	`make replay`	integer minutes, GiB
last step at $\Delta_\sigma \ge 0.74$ (`CONV-STEP`)	$< 10^{-7}$ (7.3×10^{-8})	P	$N = 18$, 20	`make replay`	$< 10^{-7}$
last steps at $0.518 \le \Delta_\sigma \le 0.625$ (`MONO-STEPS`)	$(2.2\text{–}6.1) \times 10^{-4}$	P	$N = 18$, 20	`make replay`	2 significant digits
monotone rise there (`MONO-RISE`)	every step > 0, $N = 8$–20	P	$N = 8$–20	`make replay`	sign

result	value	pkg	records	command	tolerance
shrink per step there (`MONO-SHRINK`)	8 to 19%	P	$N = 16$–20	`make replay`	integer percent
no shrink at 0.65, 0.675 (`NO-SHRINK`)	$d_{20}/d_{18} = 0.99$, 1.23	P	$N = 16$–20	`make replay`	ratio > 0.99
short-range end (`SR-L20`)	1.41924, rising	P	`grid20sri`, $N = 20$	`make replay`	5 decimals
distance to the 3d Ising value (`SR-GAP`)	6.6×10^{-3}	P	same; Ising value 1.41262528 [3]	`make replay`	2 significant digits
resummed ϵ series at $\Delta_\sigma = 0.7$ (`BP-0.7`)	1.47214(3)	P	C_0–C_7(3) (GFF folder)	`make replay` (`public/bp_x07.py`)	median to 10^{-7}; half-spread 1 digit
$N = 20$ below it and moving away (`BP-DEV`)	2.2×10^{-4}	P	`grid20`, $N = 20$; `grid18`, $N = 16$, 18; at 0.7	`make replay`	2 significant digits; order
$C_4(N = 20)/C_4(3)$ (`C4-RATIO`)	60%	P	`HT_Pert/results/HT_3d/SUMMARY.md`, row 20 consistent	`make replay` (SKIP without `HT_Pert/`)	integer percent
independent solver, tuned (`INDEP-TUNED`)	2×10^{-11} ($N \le 12$)	I	`results/cmp_*.json` (23)	`make replay`	1 significant digit
independent solver, fixed couplings (`INDEP-FIXED`)	$\le 1.8 \times 10^{-12}$	I	same	`make replay`	$\le 3 \times 10^{-12}$
independent tunings converged (`INDEP-RESID`)	$\le 10^{-12}$	I	same	`make replay`	$\le 10^{-12}$
compared values are the records (`INDEP-PKG`)	exact	I	`cmp_*.json`, `indep_extract.json`, C	`make replay`	exact
$J = 0$ counts 3, 10, 33, 110 (`test`)	as stated	I	none	`make test`	exact

A Provenance and reviews

Who wrote what. Every package records its origin in `PROVENANCE.md`, and every review is summarized in `REVIEWS.md`.

$d = 1$ **solver.** `lri_dressed_phi4_hpc` went through three stages.

- The author supplied it as a matrix-free rewrite of an earlier dressed zero-mode solver (commit e637dfd), written by OpenAI GPT-6 astra.
- Claude Code sessions (Anthropic, model claude-opus-5-5) added the layer that runs it on the symmetry cluster (node packing, cross-node locks, validation records), the window correction and the descendant diagnostic (157a2c6 to e2ca07c). Codex (OpenAI, gpt-6-astra) reviewed the plan, which was approved in round 4, and inspected the code twice. All 20 findings were fixed; the last eight fixes were not re-inspected.
- The density basis was supplied by the author as a patch (33998bb). A fresh Codex inspection found two medium defects in scripts the production does not use; they were deferred.

Both supplied archives (e637dfd and the density-basis patch) were written by OpenAI GPT-6 astra via ChatGPT (model stated by the author).

$d = 1$ **analysis and records.**

- The analysis scripts, the campaigns and this note were written by Claude Code sessions.
- The blind $d = 1$ solver was written by a Claude Code subagent (Section 7).
- No Codex results review was run on the $d = 1$ finite-coupling numbers. The public-release replay found three paper values that the stated estimators did not reproduce: the $\Delta_\sigma = 0.05$ range, the bound on the weak-coupling spreads, and the basis error. It also found one value quoted beyond its reproducible precision, the ϵ-series difference at $\Delta_\sigma = 0.249$. All four were corrected on 2026-10-13.

$d = 2, 3$**.** See the provenance files of `lri_2d_window_hpc` and `lri_3d_window_hpc`.

This note. Written by the Claude Code session `phi4HT1d`. The $d = 2$ and $d = 3$ sections were written by the sessions `phi4HT2d` and `phi4HT3d` and edited here only for consistency.

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